Networking (CS4457)

main textbooks

Peterson and Davie, *Computer Networks: A Systems Approach*

- available online `https://book.asystemsapproach.org/`, also paper version
- focused on higher-level goals, not details of specific implementations

Fall and Stevens, *TCP/IP Illustrated: Volume 1, Second Edition: The Protocols*

- available via O'Reilly through library
  - link on schedule page, or search library catalog
  - make account with your UVa email
- focused on details of the Internet and related systems

coursework

weekly quizzes

- approx 4 questions/week
- mostly autograded, will manually adjust based on work/comments
- 20% of overall course grade, lowest dropped automatically

homework assignments

- mostly programming-based
- approx. weekly deadlines
- 60% of overall course grade

final exam

- last exam slot(!)
- 20% of overall course grade

due times/late policy

assignments usually due Friday @ 11:59pm

- 90% credit for submissions up to 72 hours late, no questions asked

quizzes usually due Tuesday @ 9:15am

- done online, untimed, open notes/book
- late submissions not normally accepted (may discuss during lecture)

unusual circumstances — contact me

- can often grant exceptions/alternate arrangements
- more flexibility if contacted earlier

collaboration

quizzes, homeworks are individual

yes:

- review examples in textbook/lecture/etc.
- discuss high-level concepts (without detailed psuedocode for assignment, etc.)
- lookup up syntax/standard library documentation

no:

- sharing code/detailed psuedocode/answers
- getting code/etc. from web forums, LLMs, etc.

LLM rule of thumb: can ask LLM for things you'd ask classmate for

aside: selected standard python libraries

`bisect` — search/insert within sorted list

`collections`'s `deque` — ordered list with fast insertion/deletion at both ends

`heapq` — priority queue (using a heap)

`struct` — convert between integers and bytes and vice-versa

- `array` — convert between integers and bytes in less flexible way

not intending for you to reimplement common, non-networking tasks

may grant exceptions to use other libraries/etc. for *generic* tasks

lectures

will attempt to record (but sometimes technical stuff happens)

- will link recordings from schedule page

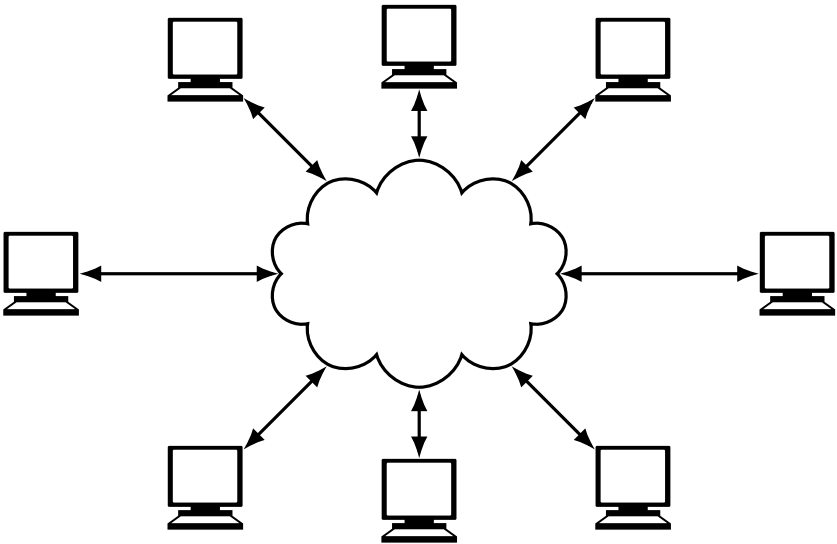
will not take attendance

homework/topic sync

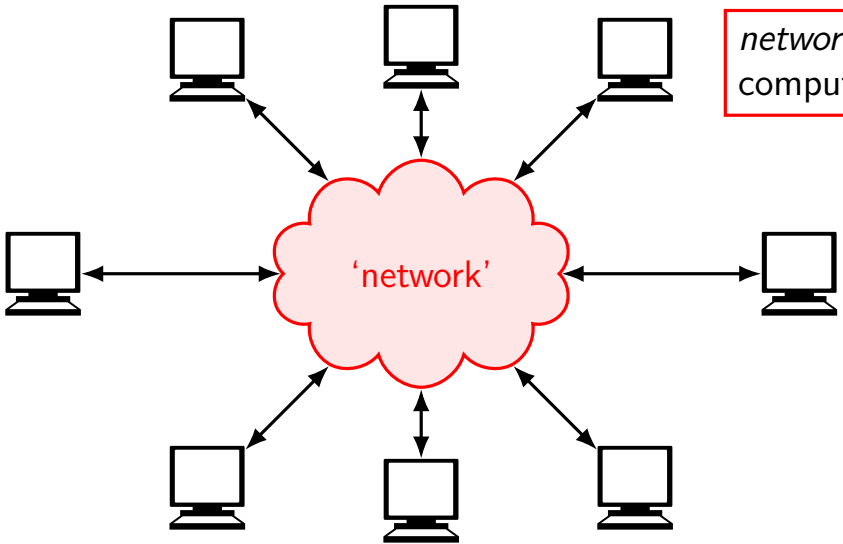
rough plan on schedule page

imagine things might get out of sync

networks / hosts aka end systems

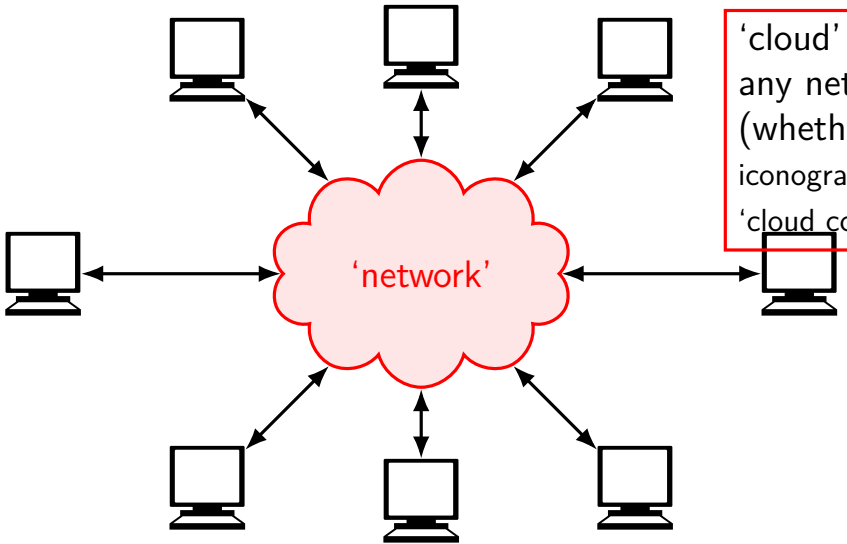


networks / hosts aka end systems



networks connect computers

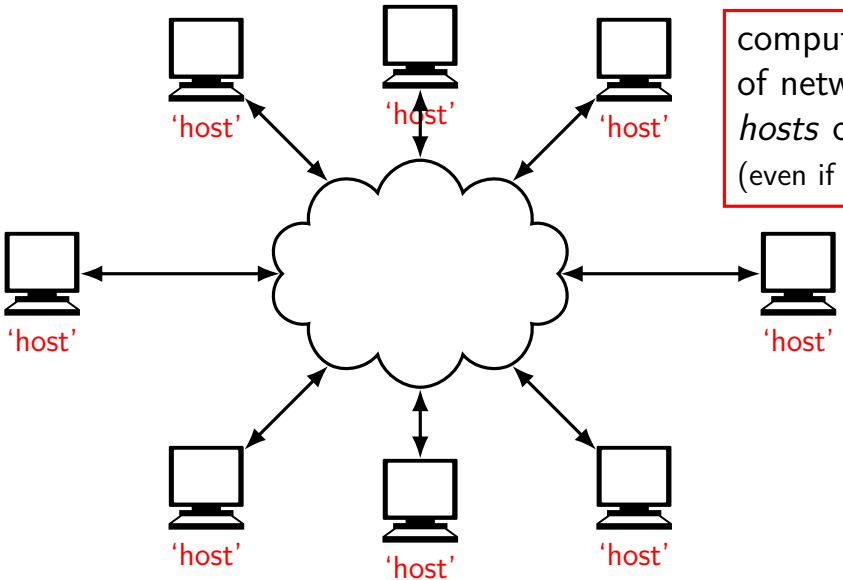
networks / hosts aka end systems



'cloud' represents any network (whether local or not)

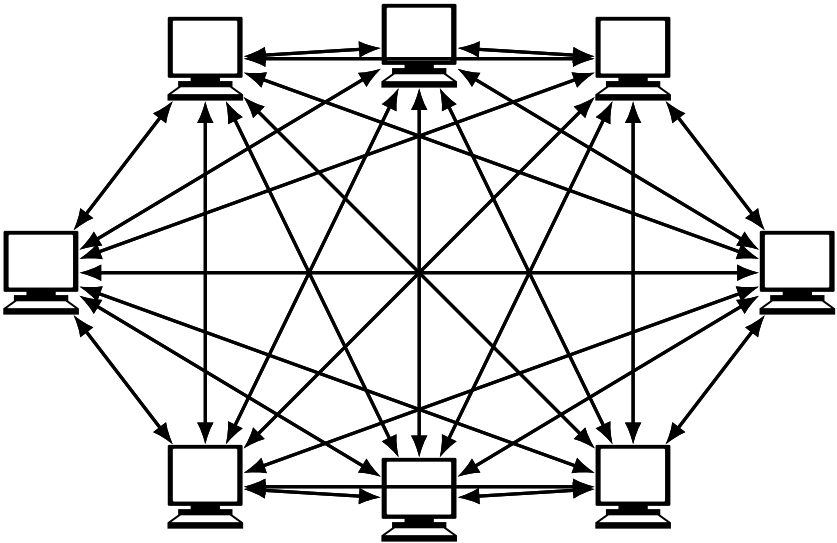
iconography predates 'cloud computing'

networks / hosts aka end systems

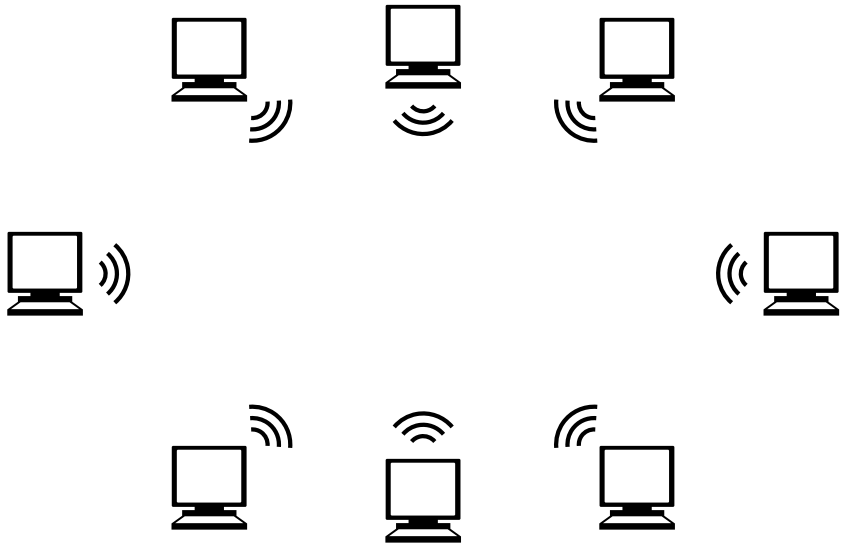


computers on edge of network called *hosts* or *end systems* (even if also 'servers')

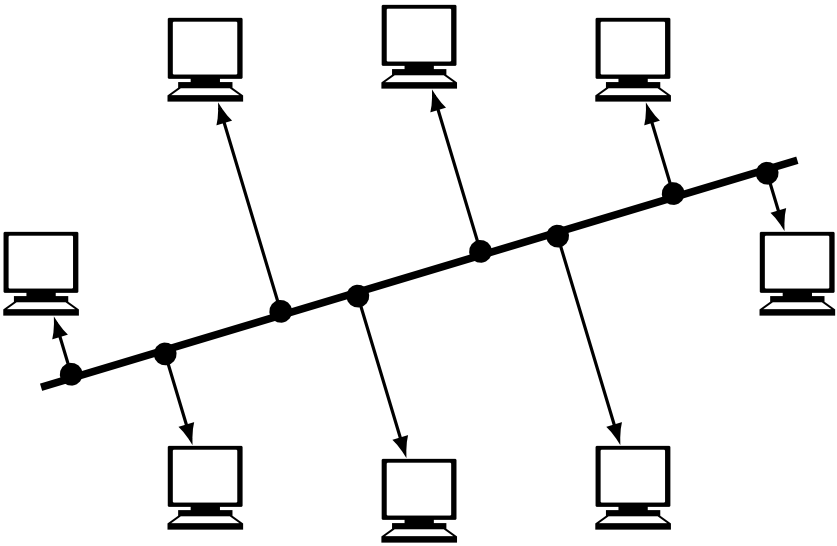
direct connections?



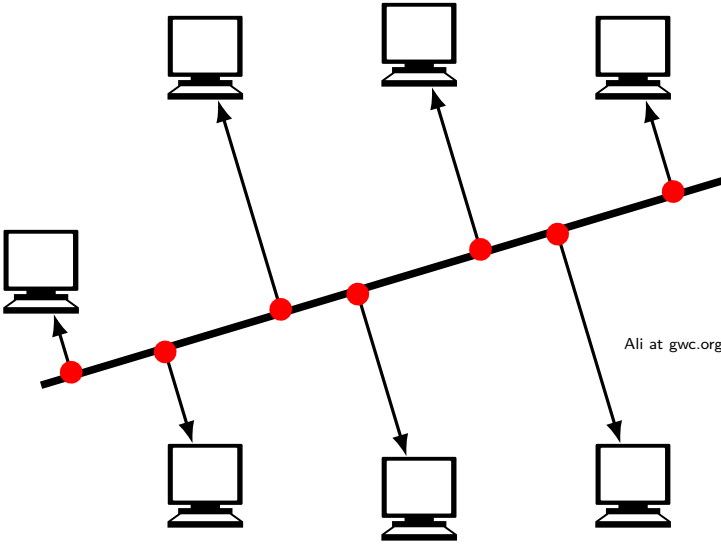
shared medium: radio?

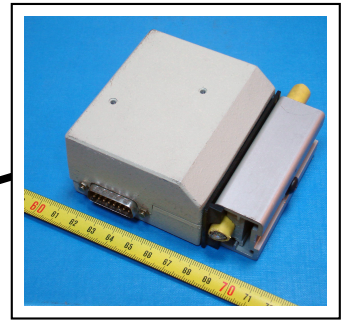


shared medium: wires



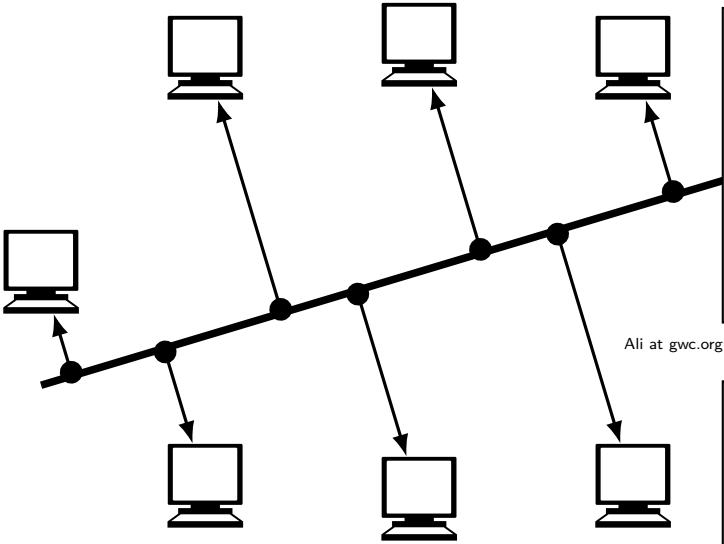
shared medium: wires

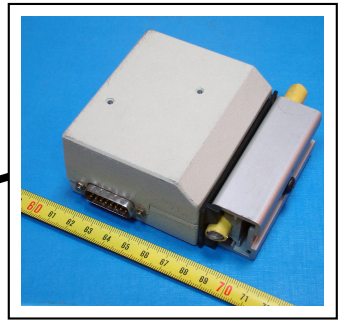




Ali at gwc.org.uk / Alistair1978 via Wikimedia commons / CC-BY-SA 2.5

shared medium: wires

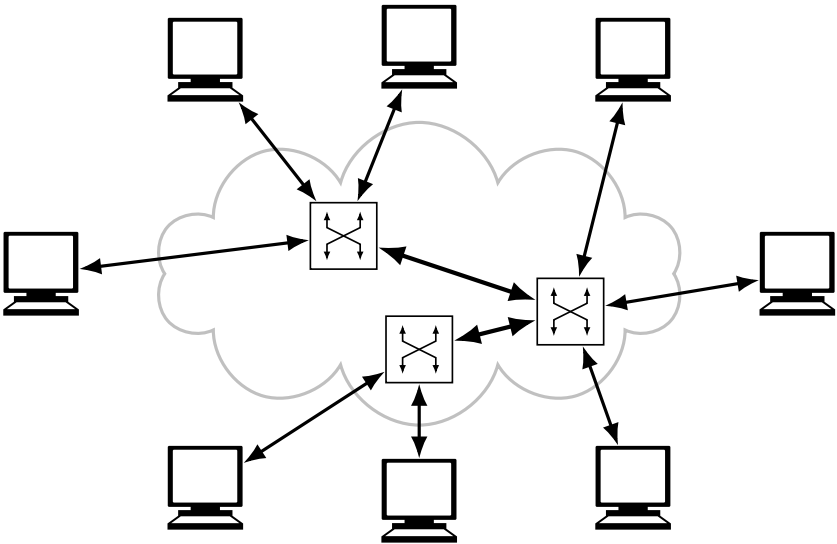




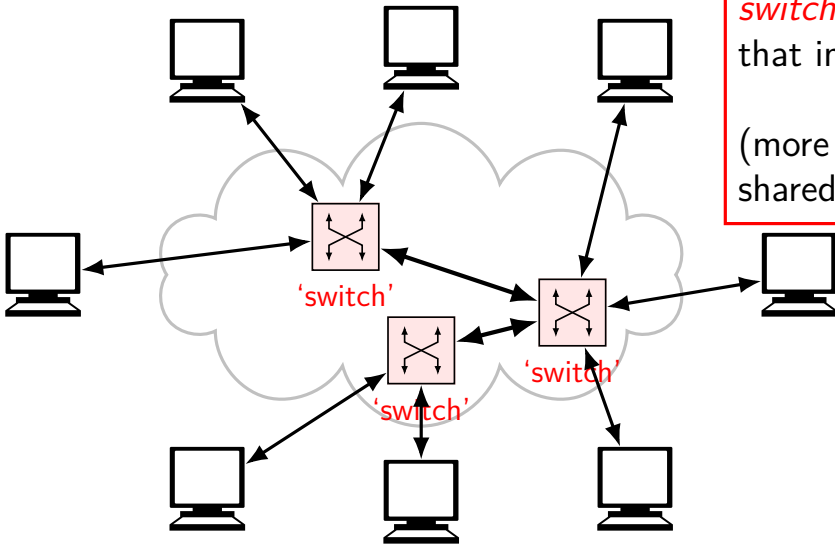
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switches / nodes / links



switches / nodes / links

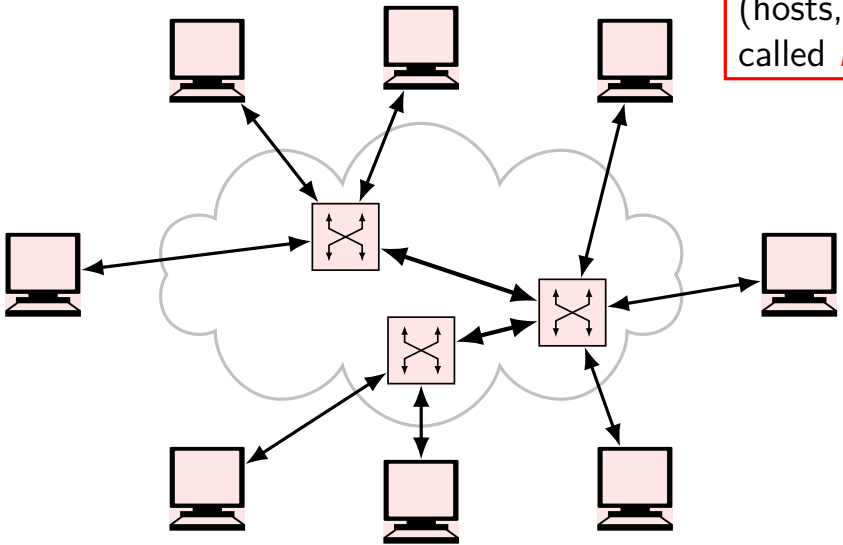


hosts directly connected to *switches*
that implement network

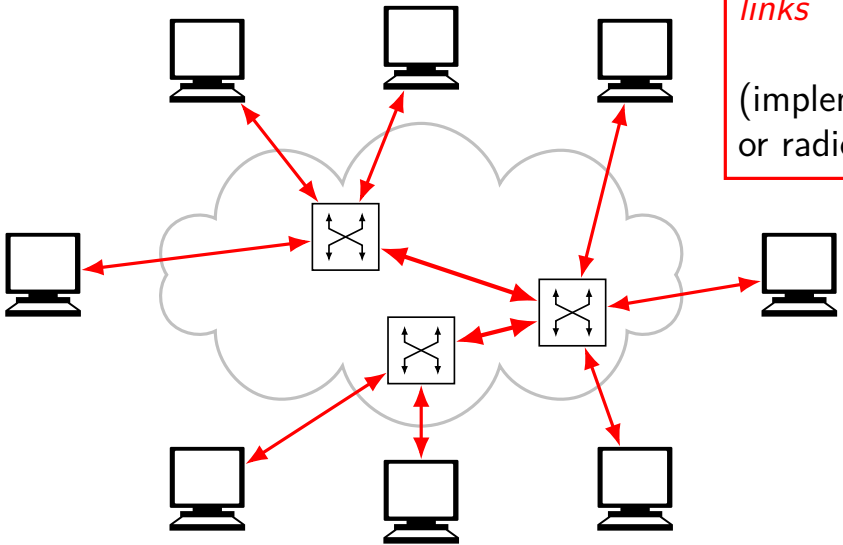
(more efficiently than shared medium)

switches / nodes / links

machines on network (hosts, switches, ...) called *nodes*



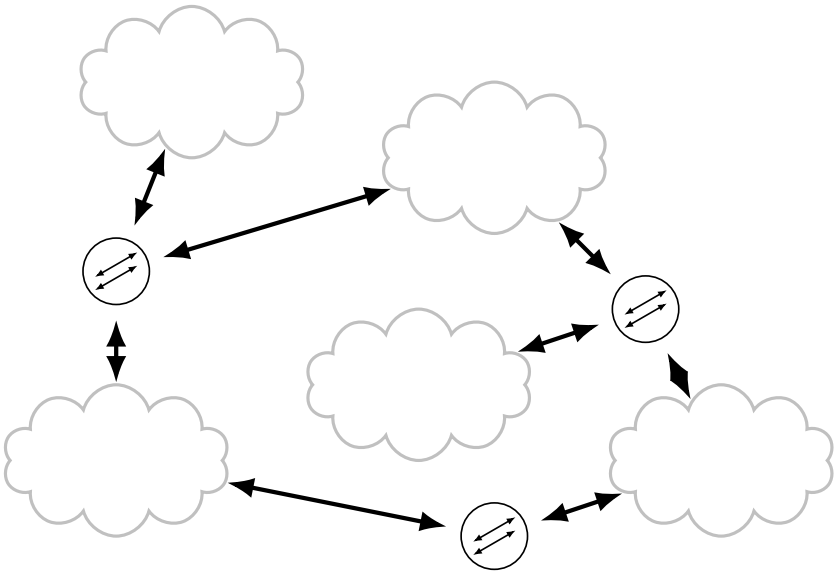
switches / nodes / links



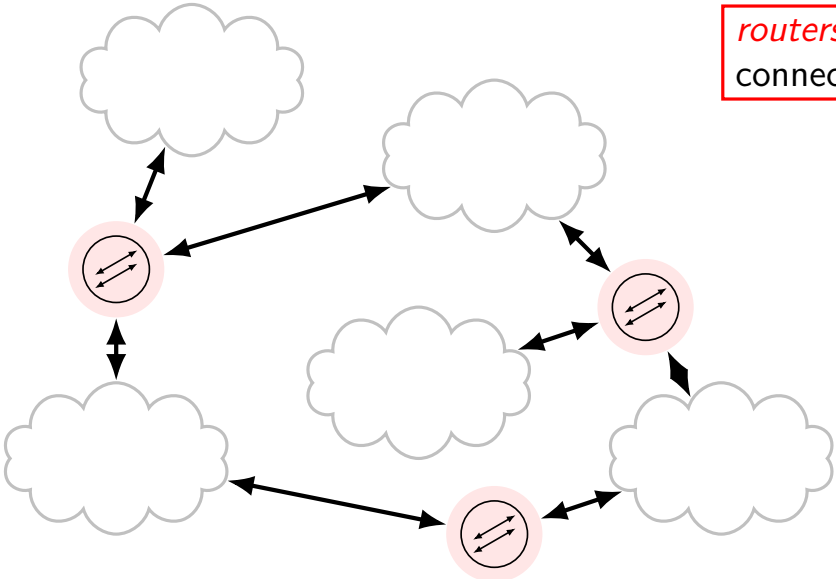
nodes connected by *links*

(implemented by wires or radio or ...)

routers / internetwork

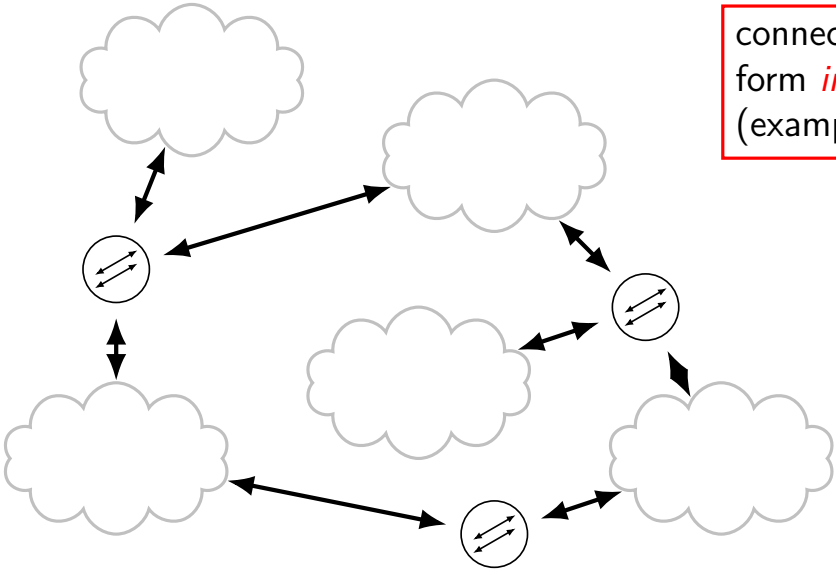


routers / internetwork



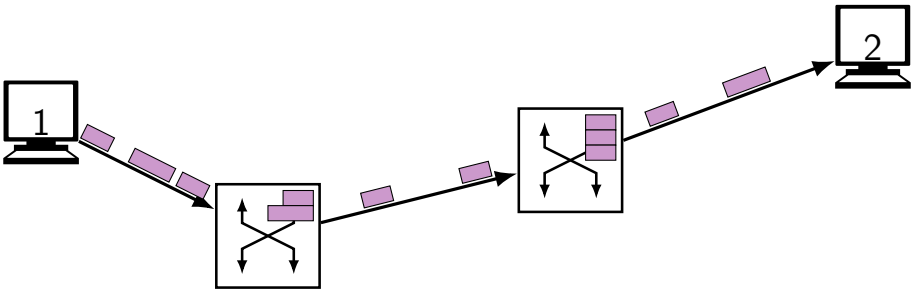
routers or *gateways* connect networks

routers / internetwork



connected networks
form *internetwork*
(example: the Internet)

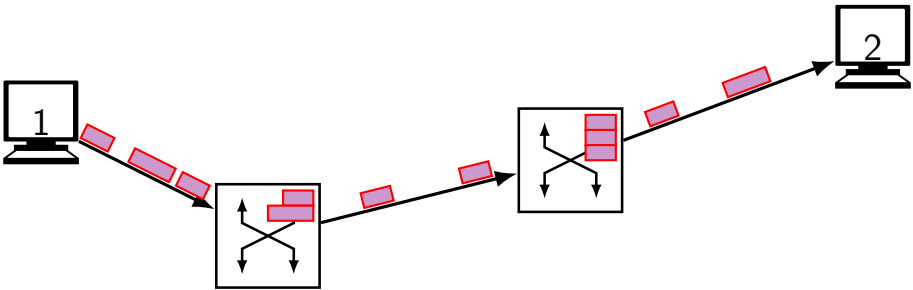
flows / packets



flow of data between two machines

flow is very general term
will depend on context how it relates to connections, sockets, etc.

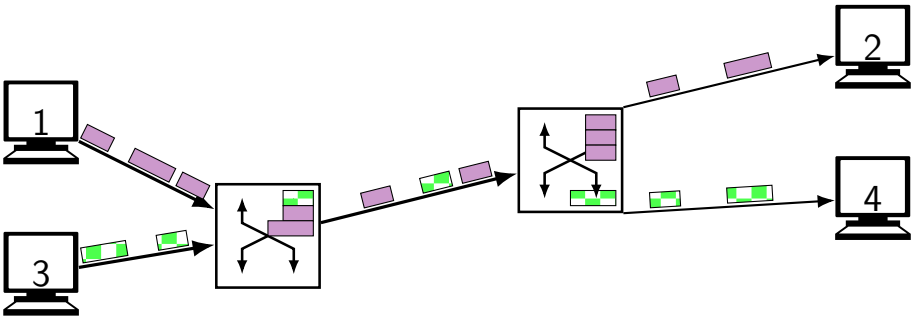
flows / packets



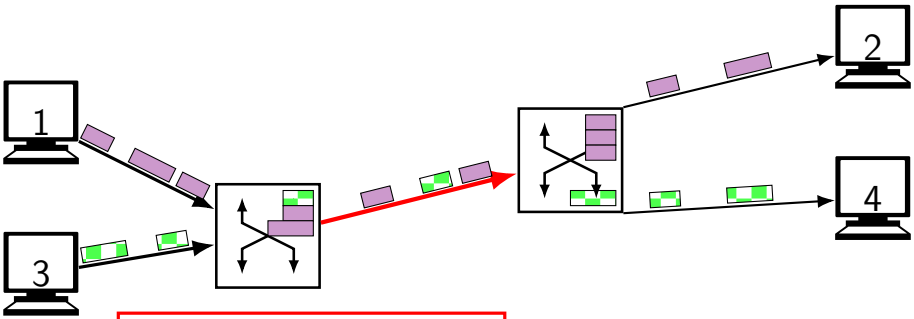
flow of data between two machines

possibly divided up into pieces,
called *packets*, *frames*, *segments*
(which name is best depends on context)

(de)multiplexing

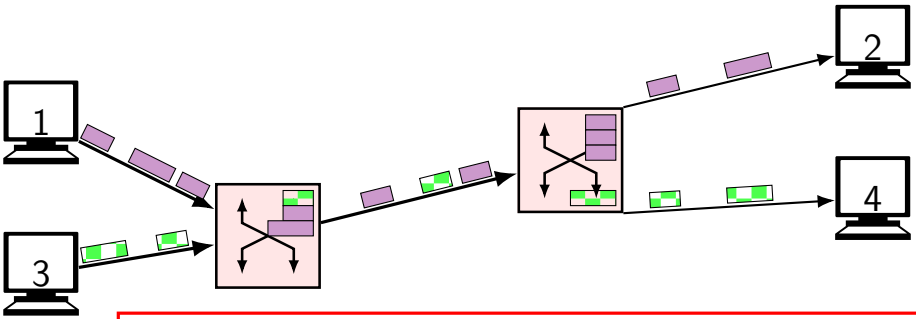


(de)multiplexing



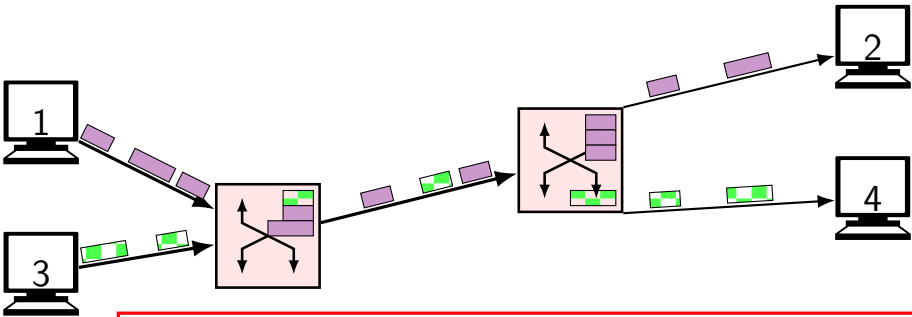
two or more flows can
share one or more links

(de)multiplexing



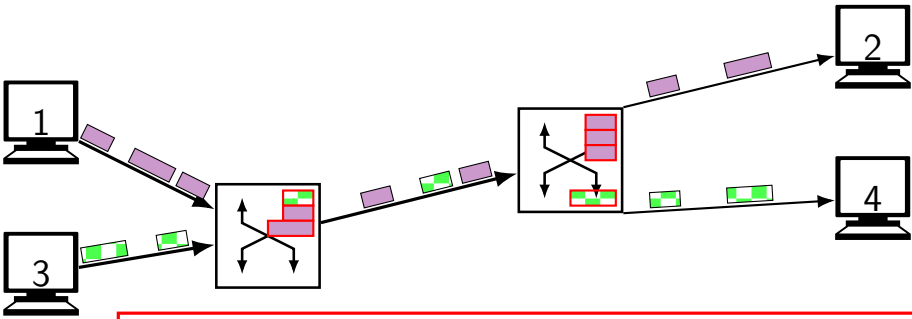
left switch *multiplexes* the two flows onto one link
right switch *demultiplexes* them to separate them

(de)multiplexing



this picture: multiplexed by dividing up *time* on link

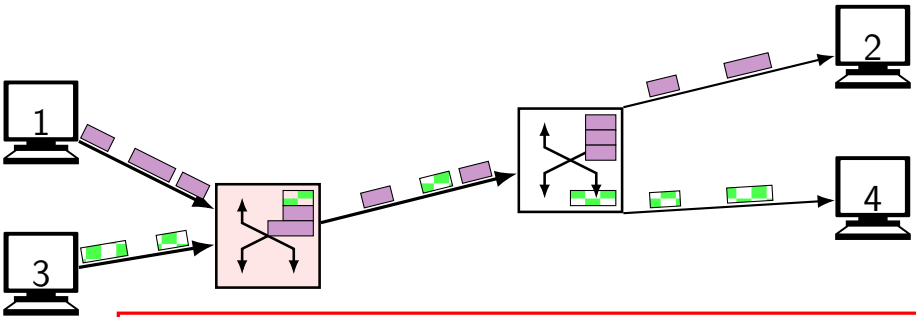
(de)multiplexing



switches usually have *buffers* (also called *queues*) hold waiting packets

absorbs temporary "bursts" where packets come faster than outgoing link can handle

(de)multiplexing

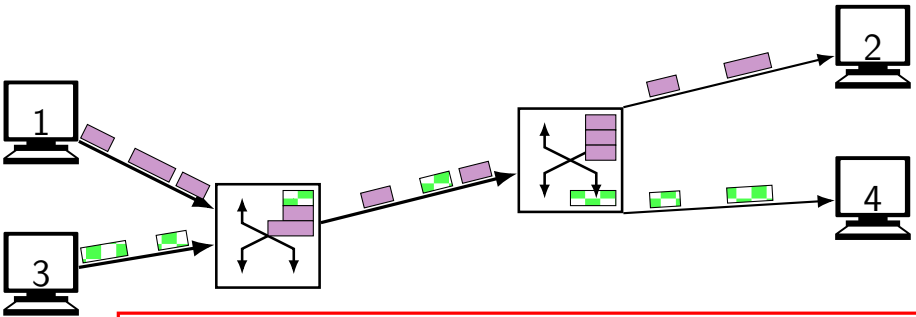


incomplete list of causes of ‘bursts’:

multiple unsynchronized flows

fast links produce packets faster for slow can send

(de)multiplexing

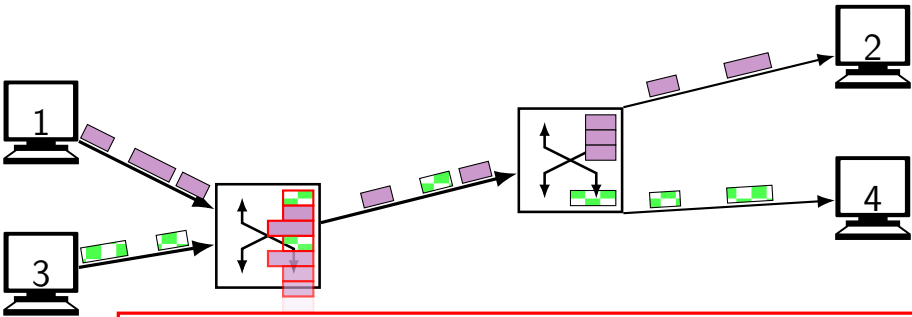


incomplete list of causes of ‘bursts’:

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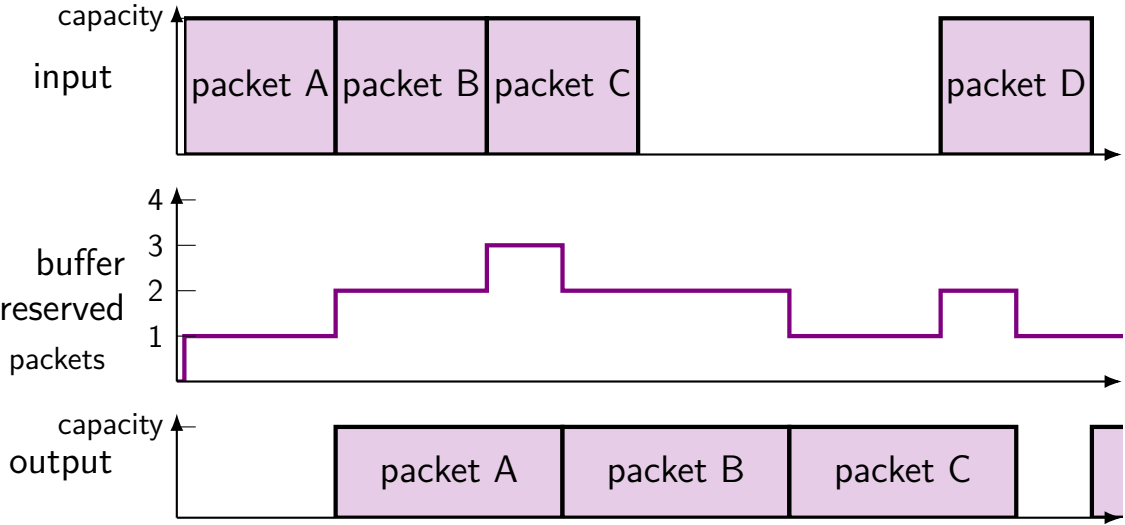
(de)multiplexing



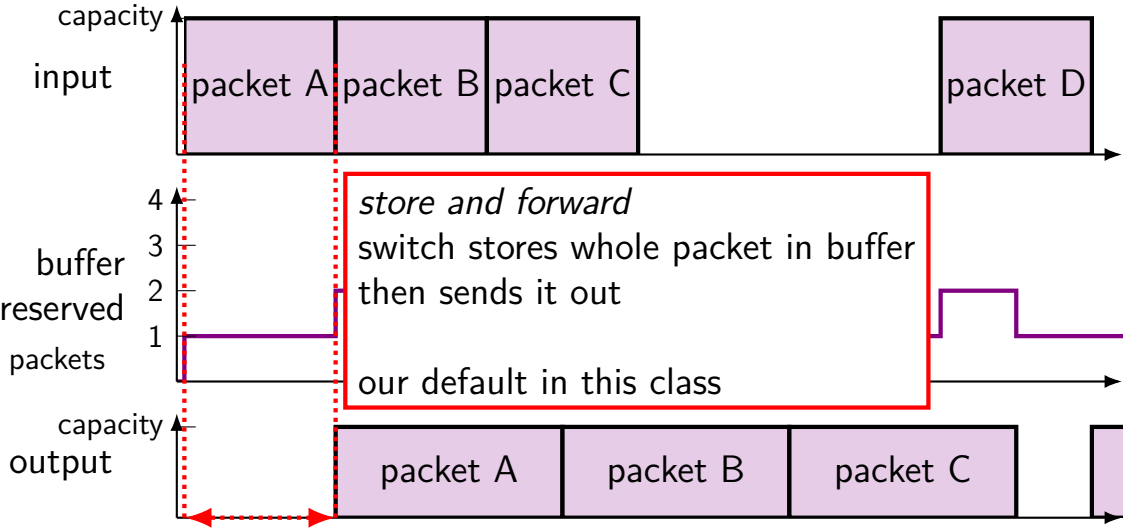
if buffer full, switch must *drop* packets
will happen eventually if overall rate faster than outgoing link

scenario is called *congestion*

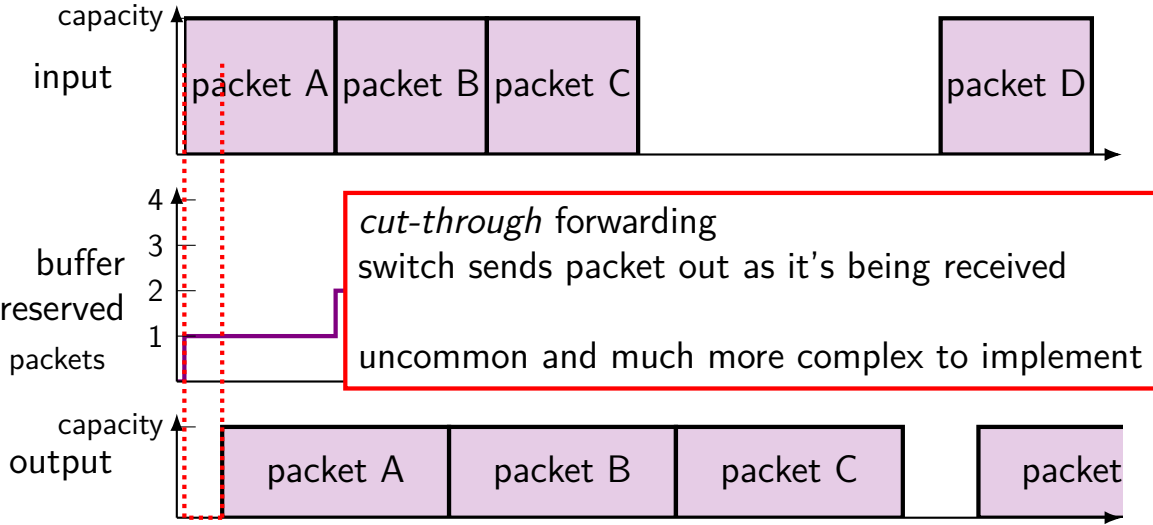
buffer usage: fast to slow, store + forward



buffer usage: fast to slow, store + forward



buffer usage: fast to slow, cut-through



channel abstractions

want to avoid custom network for each application

but applications have different needs

$\rightarrow$ multiple application interfaces to networks

common implementation of common patterns

some abstractions

stream

- continuous stream of bytes from one program to another
- 'connection' from one program to another

datagrams

- send small messages (*datagrams*)
- each datagram's destination independently set

remote procedure calls

- make function calls that run on remote machine

remote memory access

- read/write bytes of data in remote memory

...

some abstractions

stream

- continuous stream of bytes from one program to another
- 'connection' from one program to another

datagrams

- send small messages (*datagrams*)
- each datagram's destination independently set

remote procedure calls

- make function calls that run on remote machine

remote memory access

- read/write bytes of data in remote memory

...

focus on streams

this class: focus on implementing *streams of bytes*

why?

- most commonly used by applications on the Internet
- many common tasks with other abstractions

IETF

IETF = Internet Engineering Task Force

- part of non-profit called *Internet Society*

most common internet protocols standardized by IETF

most IETF documents called RFCs

- requests for comment
- have unique number

`https://rfc-editor.org`

other standard orgs

Bluetooth Special Interest Group

IEEE (Institute of Electrical and Electronics Engineers)

- Wifi, Ethernet, ...

3GPP (3rd Generation Partnership Project)

- cellular phone networks

SCTE (Society of Cable Televsion Engineers)

ITU (International Telecommunication Union)

ISO (International Organization for Standardization)

some challenges for streams

separating data into pieces network can handle

putting pieces back together

getting network to send piece to correct remote network

getting network to send piece to correct machine

getting machine to send data to correct program

getting pieces into format wires/radio/fiber/etc. can handle

handling transmission errors

some challenges for streams

separating data into pieces network can handle

putting pieces back together

getting network to send piece to correct remote network

getting ne

lots of work! don't want to implement all at once!

getting machine to send data to correct program

getting pieces into format wires/radio/fiber/etc. can handle

handling transmission errors

some challenges for streams

separating data into pieces network can handle

putting pieces in order

some parts need to be different for different local networks

getting network to send piece to correct remote network

getting network to send piece to correct machine

getting machine to send data to correct program

getting pieces into format wires/radio/fiber/etc. can handle

handling transmission errors

some challenges for streams

some parts should not concern local network implementors

separa

putting pieces back together

getting network to send piece to correct remote network

getting network to send piece to correct machine

getting machine to send data to correct program

getting pieces into format wires/radio/fiber/etc. can handle

handling transmission errors

some challenges for streams

separating

some parts should be same for different abstraction

putting pieces back together

getting network to send piece to correct remote network

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getting machine to send data to correct program

getting pieces into format wires/radio/fiber/etc. can handle

handling transmission errors

layered model

networking implemented in ‘layers’

upper layers implemented by making calls to lower layers

example: network implements ‘send data to (remote) machine’ function (“network layer”)

stream implementation calls this to implement ‘send stream to remote application’

OSI model

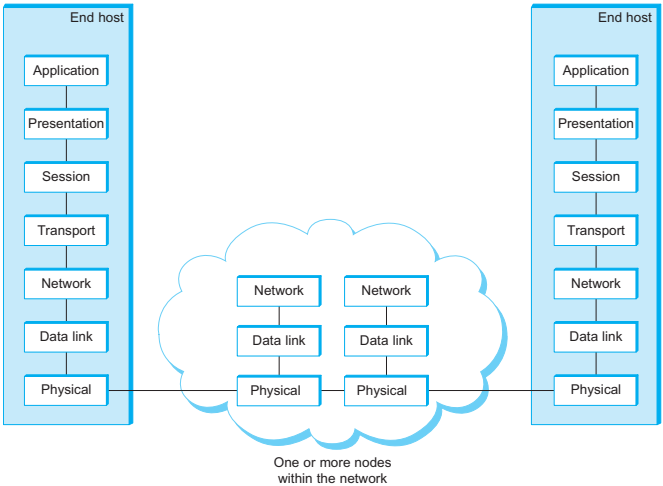
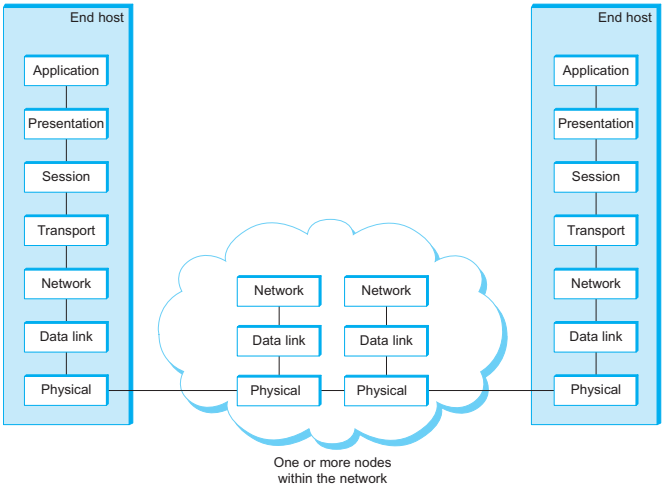


Figure 13 of Chapter 1 of Computer Networks: A Systems Approach (6th ed) (Peterson and Davie)

OSI model

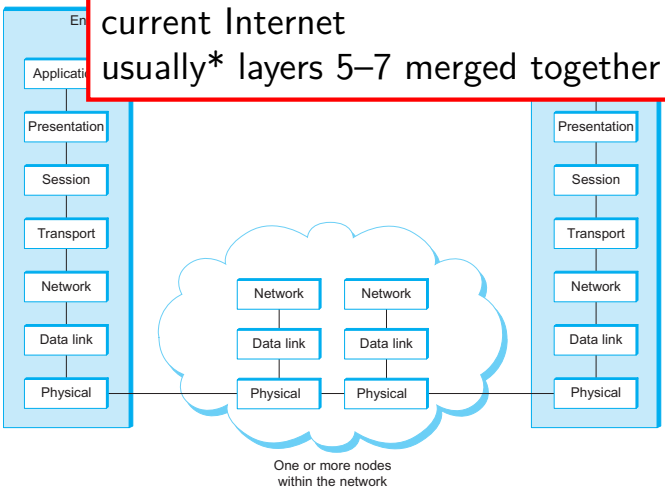


(7) application:
what requests/etc.
(6) presentation:
data format
(5) session:
manage group of streams
(4) transport:
streams of data
(3) network:
message to correct network
(2) data link:
message $\rightarrow$ bits
message to correct machine
(1) physical:
send bits/...

Figure 13 of Chapter 1 of Computer Networks: A Systems Approach (6th ed) (Peterson and Davie)

OSI model

current Internet
usually* layers 5–7 merged together



(7) application:
what requests/etc.
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Figure 13 of Chapter 1 of Computer Networks: A Systems Approach (6th ed) (Peterson and Davie)

OSI model

standardized by ISO (International Standards Organization) and ITU (International Telecommunications Union)

full set of protocols...

- file transfer, message sending, directory lookups ...

that were often implemented and sometimes used...

but mostly lost out to IETF-standardized Internet protocols

- Internet Engineering Task Force

OSI influence (1)

term ‘layer 7’, ‘layer 4’, ‘layer 3’, etc. almost always refer to OSI model

...even though most of Internet does not follow it

- early Internet protocols predate OSI

OSI influence (2)

are a lot of Internet protocols influenced by OSI protocols

OSI's DAP (directory access protocol)
- adapted into IETF's LDAP (lightweight directory access protocol)

OSI presentation layer ASN.1 used in...
- telephony (between telephone companies)
- inter-bank messaging
- lots of cryptography-related protocols
- ...

OSI's routing protocol IS-IS still common in large Internet-connected networks
- (adapted to work alongside IETF protocols)

Internet layers

OSI layer	name	examples	purpose
7	application	HTTP, SSH, SMTP, DNS, ...	application-defined meanings
4	transport	TCP, UDP, ...	reach correct program, reliablity/streams
3	network	IPv4, IPv6, ...	reach correct machine (across networks)
2	link	Ethernet, Wi-Fi, ...	coordinate shared wire/radio
1	physical	...	encode bits for wire/radio

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Internet protocols and layers (non-exhaustive)

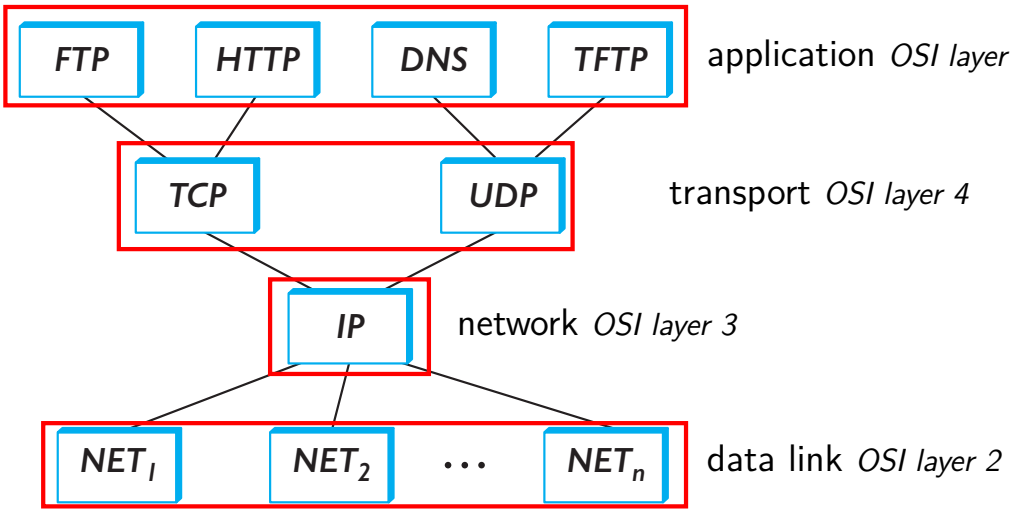


Figure 14 of Chapter 1 of Computer Networks: A Systems Approach (6th ed) (Peterson and Davie)

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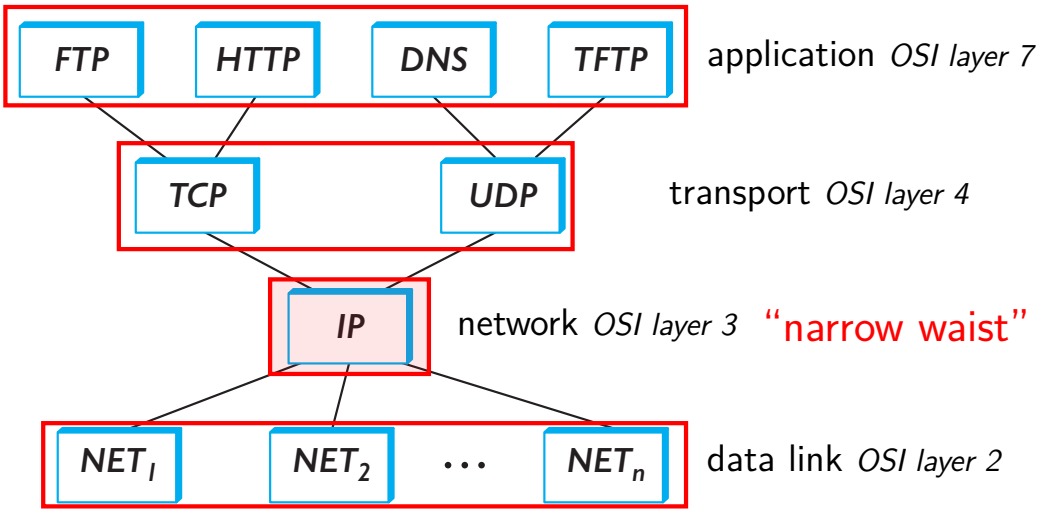


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fuzzy layers (1)

ICMP (Internet Control Message Protocol)...

implemented using a network layer...

- so seems like a transport layer protocol?

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- routing is the network layer's job
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implemented using a network layer...

- so seems like a transport layer protocol?

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- routing is the network layer's job
- so ICMP is part of network layer?

I think saying network layer is probably better...

but we're not going to be picky about it

fuzzy layers (2)

TLS (Transport Control Protocol)...

implemented on top of TCP...

- so seems like a application layer protocol?

fuzzy layers (2)

TLS (Transport Control Protocol)...

implemented on top of TCP...

- so seems like a application layer protocol?

used to send other application layer protocols

- so maybe a transport layer?
- or presentation layer?

I'll call it an application layer...

‘extra’ layers

layer terminology doesn’t always work cleanly

- often “extra” layers in practice

e.g. HTTPS:

- HTTP (app layer) on TLS (another app layer) on TCP (network) on ...

e.g. DNS over HTTPS:

- DNS (app layer) on HTTP on on TLS on TCP on ...

e.g. SFTP:

- SFTP (app layer??) on SSH (another app layer) on TCP on ...

e.g. HTTP over OpenVPN:

- HTTP on TCP on IP on OpenVPN on UDP on different IP on ...

‘extra’ layers

layer terminology doesn’t always work cleanly

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e.g. HTTPS:

- HTTP (app layer) on TLS (another app layer) on TCP (network) on ...

e.g. DNS over HTTPS:

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e.g. SFTP:

- SFTP (app layer??) on SSH (another app layer) on TCP on ...

e.g. HTTP over OpenVPN:

- HTTP on TCP on IP on OpenVPN on UDP on different IP on ...

protocols usually over HTTP

SOAP (Simple Object Access Protocol) — messaging/remote procedure calls

gRPC (originally form Google) — remote procedure calls

HLS (HTTP Live Streaming) — video streaming

DASH (Dynamic Adaptive Streaming over HTTP) — video streaming

...

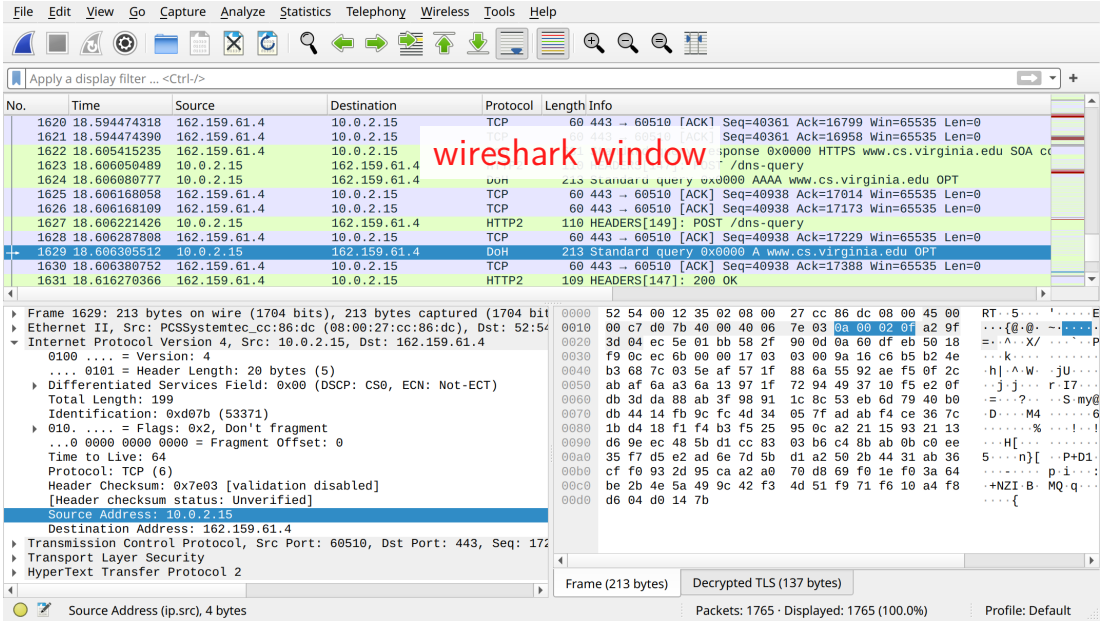
packet capture tools

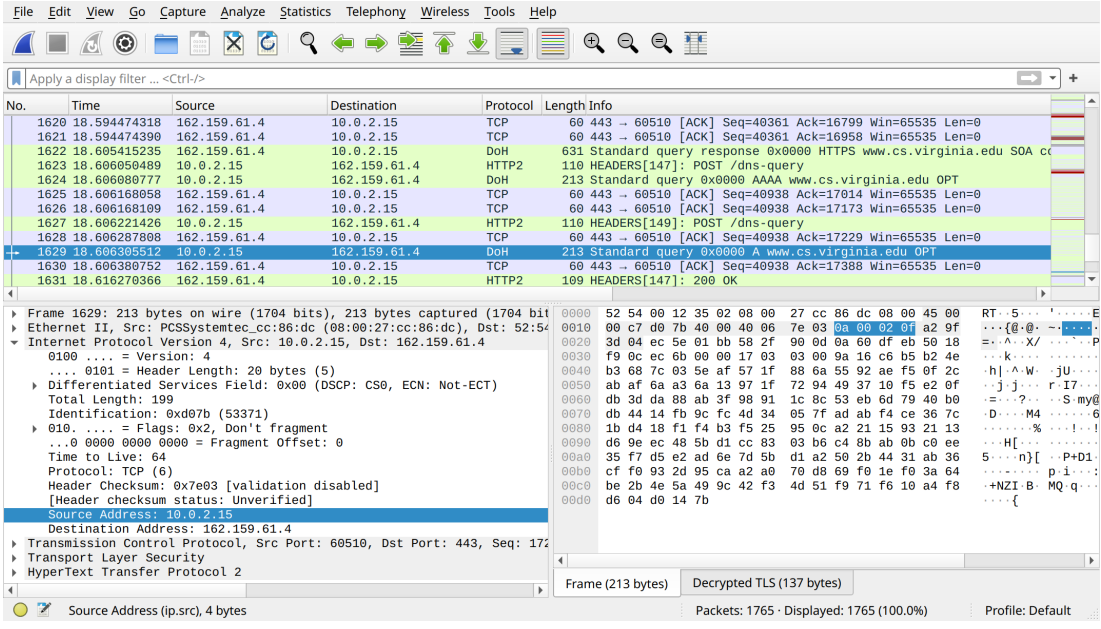
packet capture = log of everything sent/received on some link(s)

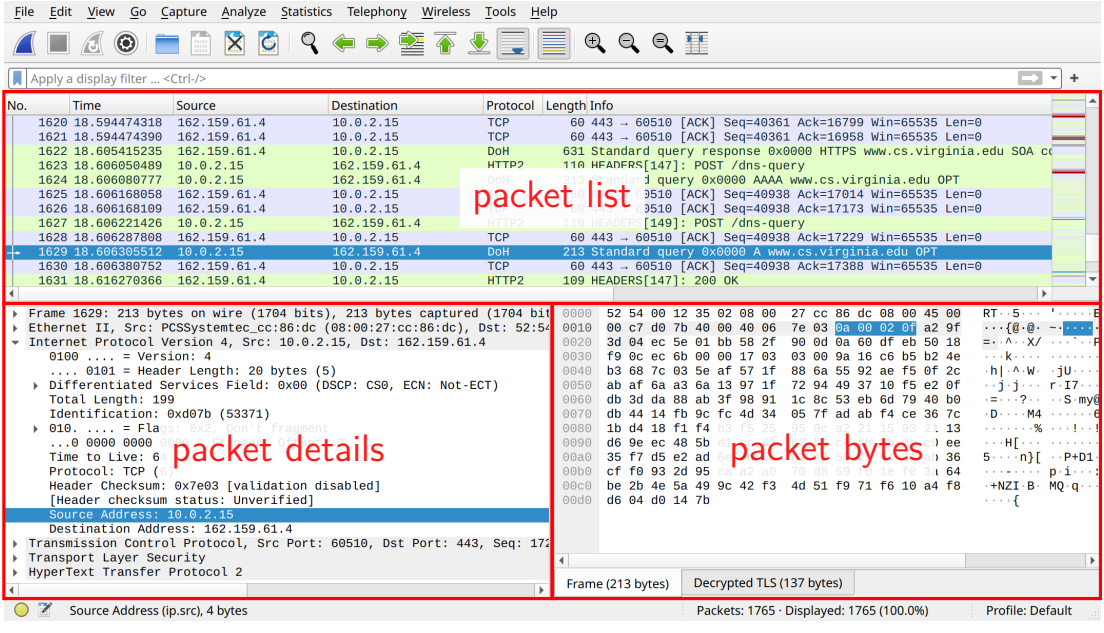
wireshark is popular tool for making, analyzing packet captures

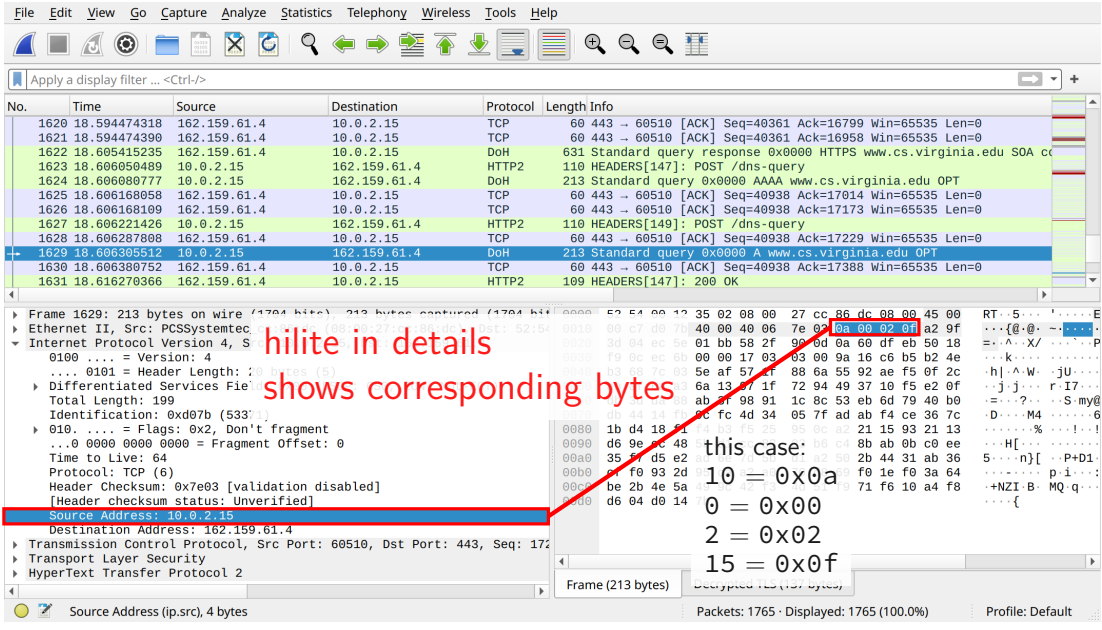
will be showing screenshots from that

you can download these packet captures, follow along in wireshark


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1620 18.594474318 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16799 Win=65535 Len=0
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1622 18.605415235 162.159.61.4 10.0.2.15 ponse 0x0000 HTTPS www.cs.virginia.edu SOA c
1623 18.606050489 10.0.2.15 162.159.61.4 /dns-query
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1625 18.606168058 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17014 Win=65535 Len=0
1626 18.606168109 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17173 Win=65535 Len=0
1627 18.606221426 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[149]: POST /dns-query
1628 18.606287808 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17229 Win=65535 Len=0
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wireshark window
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 bit
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:54
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 199
Identification: 0xd07b (53371)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x7e03 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.2.15
Destination Address: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 172
Transport Layer Security
HyperText Transfer Protocol 2
0000 52 54 00 12 35 02 08 00 27 cc 86 dc 08 00 45 00 RT··5··· '·····E
0010 00 c7 d0 7b 40 00 40 06 7e 03 0a 00 02 0f a2 9f ···{@·@· ~·····
0020 3d 04 ec 5e 01 bb 58 2f 90 0d 0a 60 df eb 50 18 =··^··X/ ···`··P
0030 f9 0c ec 6b 00 00 17 03 03 00 9a 16 c6 b5 b2 4e ···k···· ·····N
0040 b3 68 7c 03 5e af 57 1f 88 6a 55 92 ae f5 0f 2c ·h|·^·W· ·jU····,
0050 ab af 6a a3 6a 13 97 1f 72 94 49 37 10 f5 e2 0f ··j·j··· r·I7····
0060 db 3d da 88 ab 3f 98 91 1c 8c 53 eb 6d 79 40 b0 ·=···?·· ··S·my@·
0070 db 44 14 fb 9c fc 4d 34 05 7f ad ab f4 ce 36 7c ·D····M4 ·····6|
0080 1b d4 18 f1 f4 b3 f5 25 95 0c a2 21 15 93 21 13 ·······% ···!··!·
0090 d6 9e ec 48 5b d1 cc 83 03 b6 c4 8b ab 0b c0 ee ···H[··· ····
00a0 35 f7 d5 e2 ad 6e 7d 5b d1 a2 50 2b 44 31 ab 36 5····n}[··P+D1·6
00b0 cf f0 93 2d 95 ca a2 a0 70 d8 69 f0 1e f0 3a 64 ···-···· p·i···:d
00c0 be 2b 4e 5a 49 9c 42 f3 4d 51 f9 71 f6 10 a4 f8 ·+NZI·B· MQ·q····
00d0 d6 04 d0 14 7b ····{
Frame (213 bytes)
Decrypted TLS (137 bytes)
Source Address (ip.src), 4 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default


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Transport Layer Security
HyperText Transfer Protocol 2
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0010 00 c7 d0 7b 40 00 40 06 7e 03 0a 00 02 0f a2 9f ···{@·@· ~·····
0020 3d 04 ec 5e 01 bb 58 2f 90 0d 0a 60 df eb 50 18 =··^··X/ ···`··P
0030 f9 0c ec 6b 00 00 17 03 03 00 9a 16 c6 b5 b2 4e ···k···· ·····N
0040 b3 68 7c 03 5e af 57 1f 88 6a 55 92 ae f5 0f 2c ·h|·^·W· ·jU····,
0050 ab af 6a a3 6a 13 97 1f 72 94 49 37 10 f5 e2 0f ··j·j··· r·I7····
0060 db 3d da 88 ab 3f 98 91 1c 8c 53 eb 6d 79 40 b0 ·=···?·· ··S·my@·
0070 db 44 14 fb 9c fc 4d 34 05 7f ad ab f4 ce 36 7c ·D····M4 ·····6|
0080 1b d4 18 f1 f4 b3 f5 25 95 0c a2 21 15 93 21 13 ·······% ···!··!·
0090 d6 9e ec 48 5b d1 cc 83 03 b6 c4 8b ab 0b c0 ee ···H[··· ········
00a0 35 f7 d5 e2 ad 6e 7d 5b d1 a2 50 2b 44 31 ab 36 5····n}[··P+D1·6
00b0 cf f0 93 2d 95 ca a2 a0 70 d8 69 f0 1e f0 3a 64 ···-···· p·i···:d
00c0 be 2b 4e 5a 49 9c 42 f3 4d 51 f9 71 f6 10 a4 f8 ·+NZI·B· MQ·q····
00d0 d6 04 d0 14 7b ····{
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Decrypted TLS (137 bytes)
Source Address (ip.src), 4 bytes
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Total Length: 199
Identification: 0xd07b (53371)
010. .... = Fla
...0 0000 0000
Time to Live: 6
Protocol: TCP (
Header Checksum: 0x7e03 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.2.15
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Transport Layer Security
HyperText Transfer Protocol 2
packet details
0000 52 54 00 12 35 02 08 00 27 cc 86 dc 08 00 45 00 RT··5··· '·····E
0010 00 c7 d0 7b 40 00 40 06 7e 03 0a 00 02 0f a2 9f ···{@·@· ~·····
0020 3d 04 ec 5e 01 bb 58 2f 90 0d 0a 60 df eb 50 18 =··^··X/ ···`··P
0030 f9 0c ec 6b 00 00 17 03 03 00 9a 16 c6 b5 b2 4e ···k···· ·····N
0040 b3 68 7c 03 5e af 57 1f 88 6a 55 92 ae f5 0f 2c ·h|·^·W· ·jU····,
0050 ab af 6a a3 6a 13 97 1f 72 94 49 37 10 f5 e2 0f ··j·j··· r·I7····
0060 db 3d da 88 ab 3f 98 91 1c 8c 53 eb 6d 79 40 b0 ·=···?·· ··S·my@·
0070 db 44 14 fb 9c fc 4d 34 05 7f ad ab f4 ce 36 7c ·D····M4 ·····6|
0080 1b d4 18 f1 f4 13 ····· % ···!··!
0090 d6 9e ec 48 5b ···H[··· ····
00a0 35 f7 d5 e2 ad 5···n}[··P+D1··
00b0 cf f0 93 2d 95 ···-···· p·i···:
00c0 be 2b 4e 5a 49 9c 42 f3 4d 51 f9 71 f6 10 a4 f8 ·+NZI·B· MQ·q····
00d0 d6 04 d0 14 7b ····{
packet bytes
Frame (213 bytes)
Decrypted TLS (137 bytes)
Source Address (ip.src), 4 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default



hilite in details shows corresponding bytes

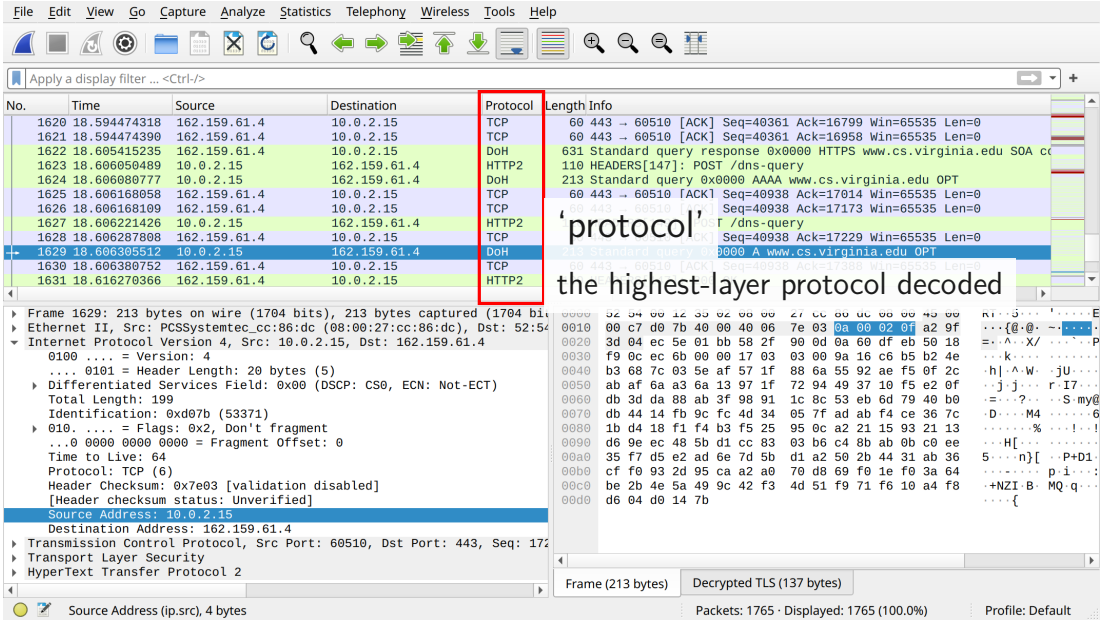
this case:

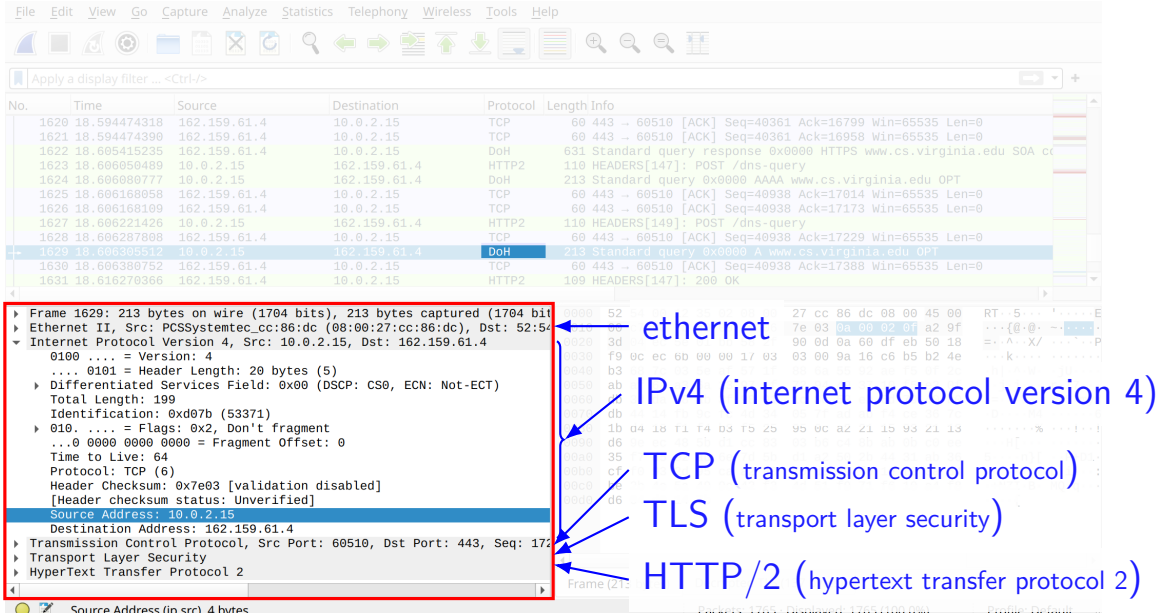
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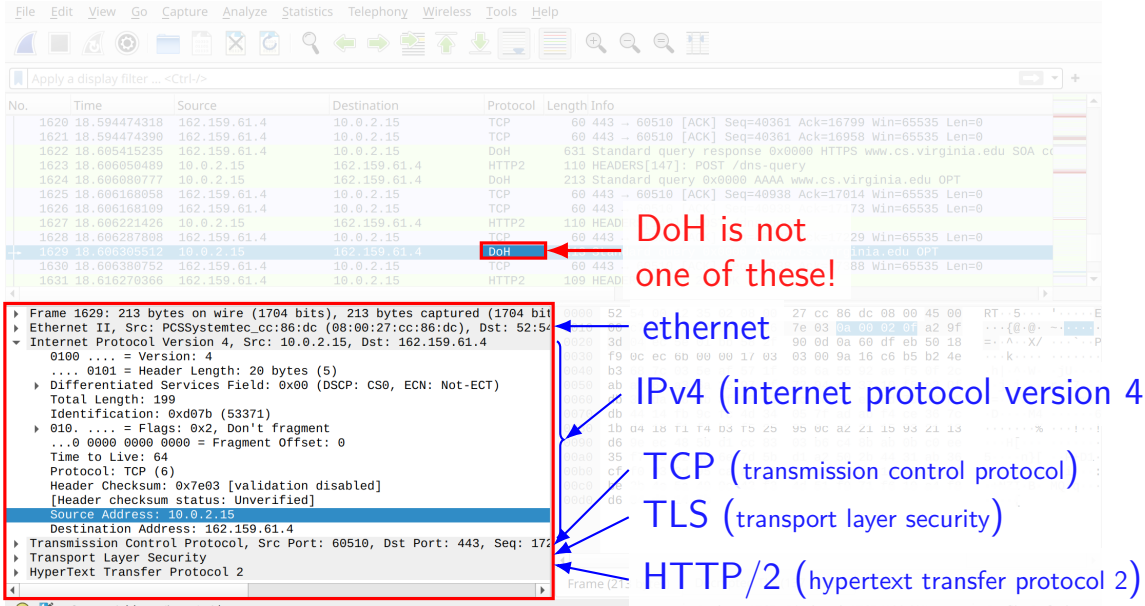
0 = 0x00

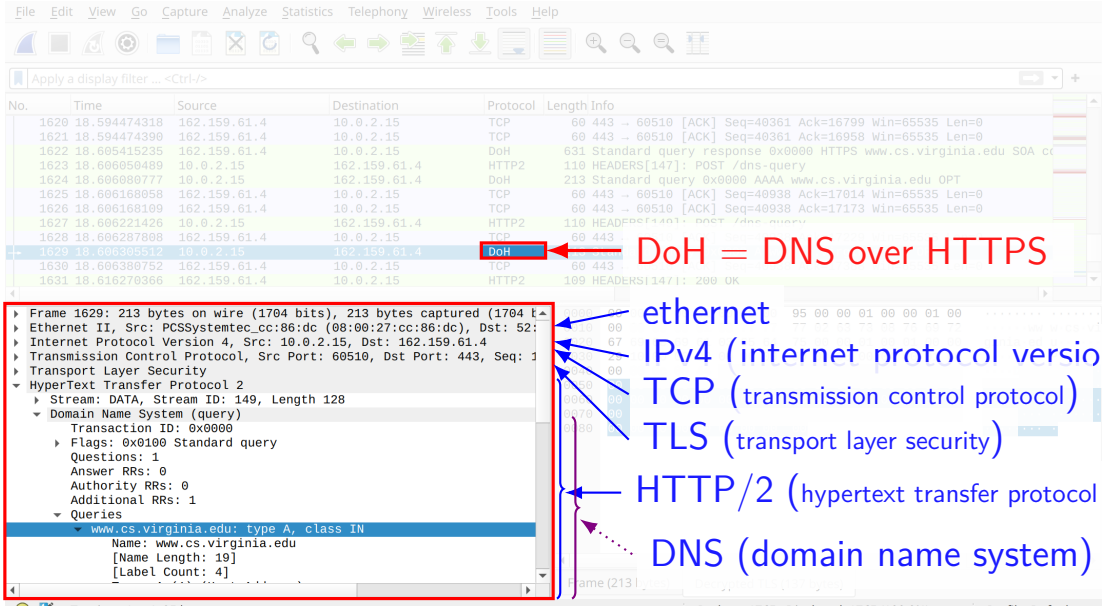
2 = 0x02

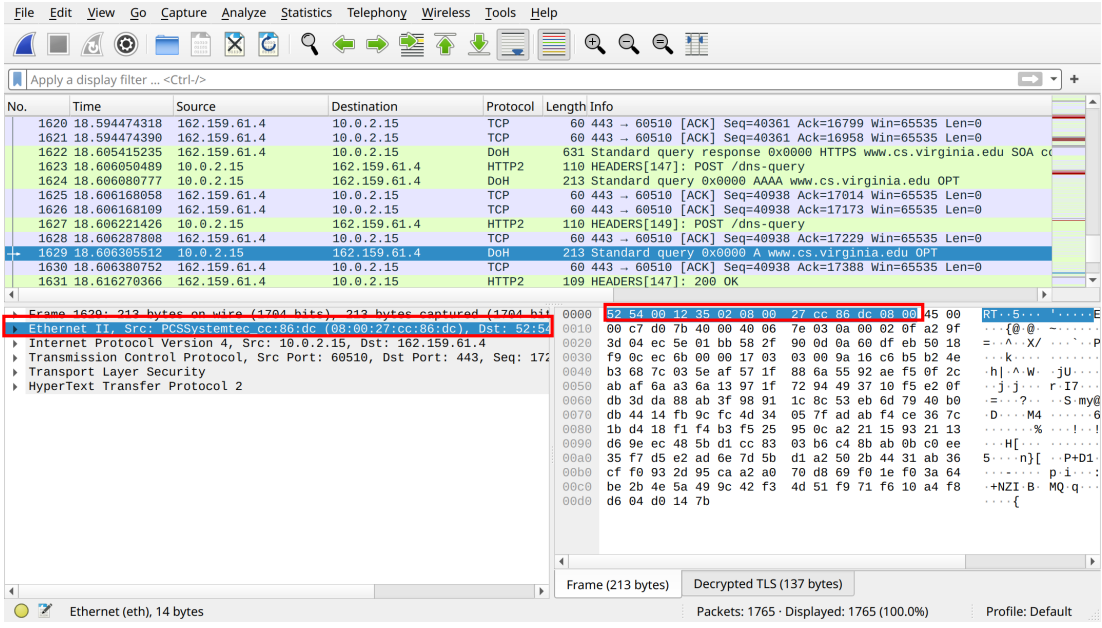
15 = 0x0f

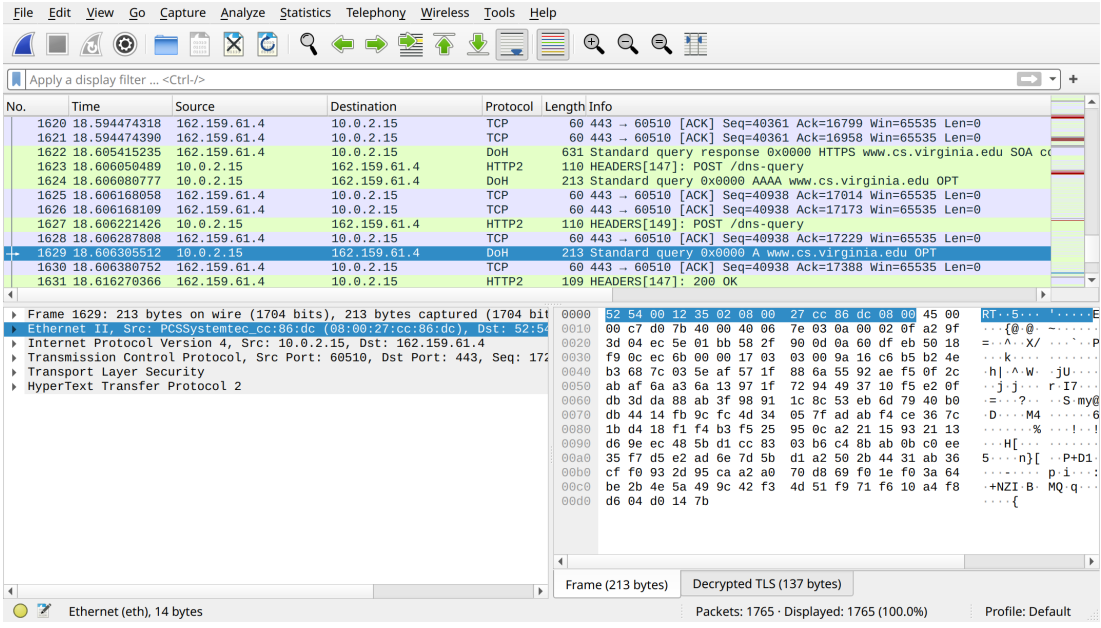

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1630 18.606380752 162.159.61.4 10.0.2.15 TCP
1631 18.616270366 162.159.61.4 10.0.2.15 HTTP2
‘protocol’
the highest-layer protocol decoded
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 bi
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:54
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010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
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0020 3d 04 ec 5e 01 bb 58 2f 90 0d 0a 60 df eb 50 18
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0050 ab af 6a a3 6a 13 97 1f 72 94 49 37 10 f5 e2 0f
0060 db 3d da 88 ab 3f 98 91 1c 8c 53 eb 6d 79 40 b0
0070 db 44 14 fb 9c fc 4d 34 05 7f ad ab f4 ce 36 7c
0080 1b d4 18 f1 f4 b3 f5 25 95 0c a2 21 15 93 21 13
0090 d6 9e ec 48 5b d1 cc 83 03 b6 c4 8b ab 0b c0 ee
00a0 35 f7 d5 e2 ad 6e 7d 5b d1 a2 50 2b 44 31 ab 36
00b0 cf f0 93 2d 95 ca a2 a0 70 d8 69 f0 1e f0 3a 64
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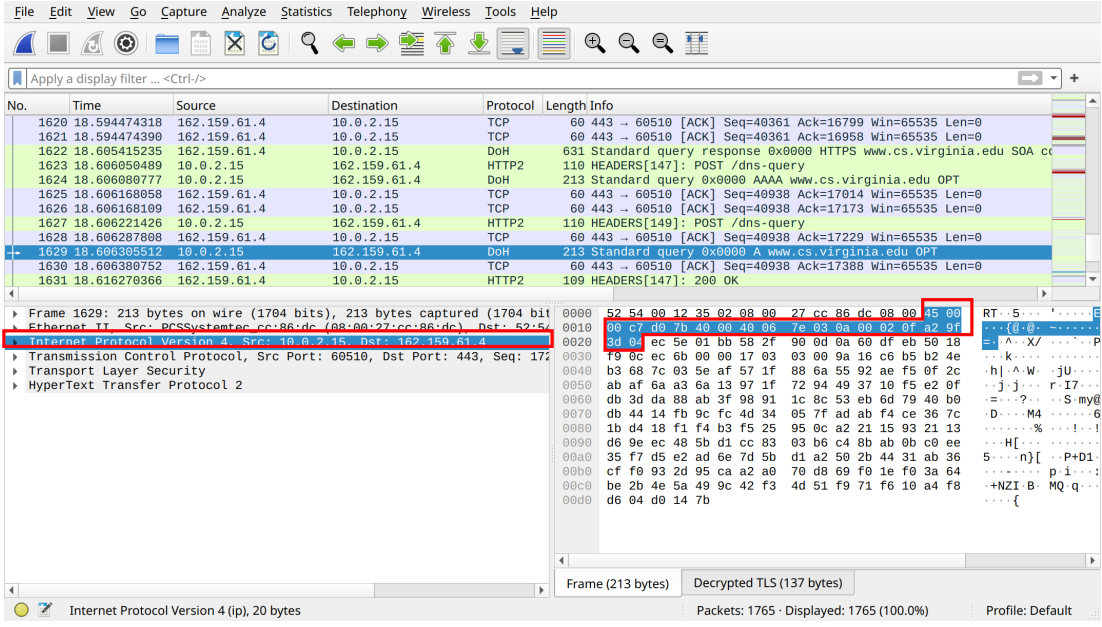

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...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x7e03 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.0.2.15
Destination Address: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 172
Transport Layer Security
HyperText Transfer Protocol 2
Source Address (ip.src), 4 bytes
ethernet
IPv4 (internet protocol version 4)
TCP (transmission control protocol)
TLS (transport layer security)
HTTP/2 (hypertext transfer protocol 2)

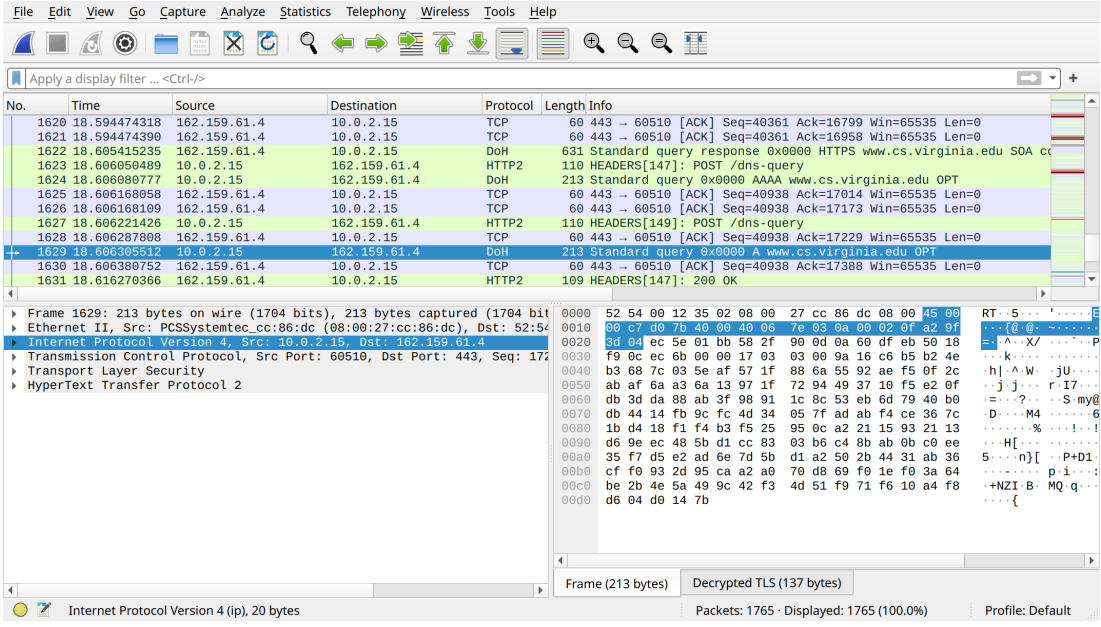

DoH is not one of these!
ethernet
IPv4 (internet protocol version 4)
TCP (transmission control protocol)
TLS (transport layer security)
HTTP/2 (hypertext transfer protocol 2)

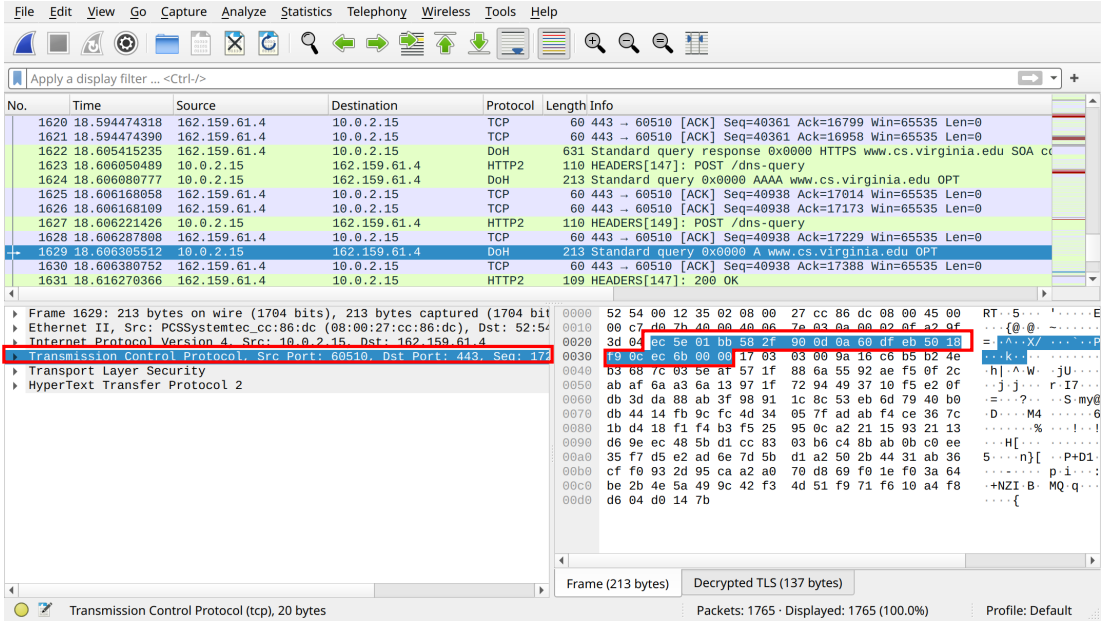

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1620 18.594474318 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16799 Win=65535 Len=0
1621 18.594474390 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16958 Win=65535 Len=0
1622 18.605415235 162.159.61.4 10.0.2.15 DoH 631 Standard query response 0x0000 HTTPS www.cs.virginia.edu SOA c
1623 18.606050489 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[147]: POST /dns-query
1624 18.606080777 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 AAAA www.cs.virginia.edu OPT
1625 18.606168058 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17014 Win=65535 Len=0
1626 18.606168109 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17173 Win=65535 Len=0
1627 18.606221426 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[149]: POST /dns-query
1628 18.606287808 162.159.61.4 10.0.2.15 TCP 60 443
1629 18.606305512 10.0.2.15 162.159.61.4 DoH
1630 18.606380752 162.159.61.4 10.0.2.15 TCP 60 443
1631 18.616270366 162.159.61.4 10.0.2.15 HTTP2 109 HEADERS[147]: 200 OK
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 b
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 1
Transport Layer Security
HyperText Transfer Protocol 2
Stream: DATA, Stream ID: 149, Length 128
Domain Name System (query)
Transaction ID: 0x0000
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
Queries
www.cs.virginia.edu: type A, class IN
Name: www.cs.virginia.edu
[Name Length: 19]
[Label Count: 4]
Text item (text), 25 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default
DoH = DNS over HTTPS
ethernet
IPv4 (internet protocol version 4)
TCP (transmission control protocol)
TLS (transport layer security)
HTTP/2 (hypertext transfer protocol 2)
DNS (domain name system)

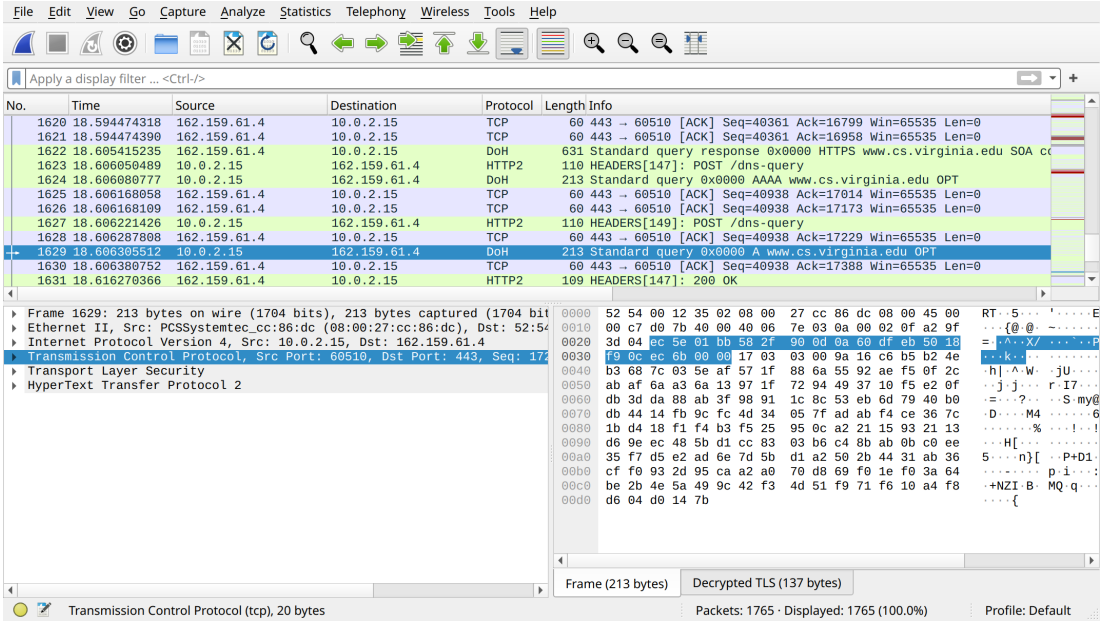

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1620 18.594474318 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16799 Win=65535 Len=0
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1622 18.605415235 162.159.61.4 10.0.2.15 DoH 631 Standard query response 0x0000 HTTPS www.cs.virginia.edu SOA c
1623 18.606050489 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[147]: POST /dns-query
1624 18.606080777 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 AAAA www.cs.virginia.edu OPT
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1626 18.606168109 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17173 Win=65535 Len=0
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1628 18.606287808 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17229 Win=65535 Len=0
1629 18.606305512 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 A www.cs.virginia.edu OPT
1630 18.606380752 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17388 Win=65535 Len=0
1631 18.616270366 162.159.61.4 10.0.2.15 HTTP2 109 HEADERS[147]: 200 OK
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 bit
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:54
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 172
Transport Layer Security
HyperText Transfer Protocol 2
0000 52 54 00 12 35 02 08 00 27 cc 86 dc 08 00 45 00
0010 00 c7 d0 7b 40 00 40 06 7e 03 0a 00 02 0f a2 9f
0020 3d 04 ec 5e 01 bb 58 2f 90 0d 0a 60 df eb 50 18
0030 f9 0c ec 6b 00 00 17 03 03 00 9a 16 c6 b5 b2 4e
0040 b3 68 7c 03 5e af 57 1f 88 6a 55 92 ae f5 0f 2c
0050 ab af 6a a3 6a 13 97 1f 72 94 49 37 10 f5 e2 0f
0060 db 3d da 88 ab 3f 98 91 1c 8c 53 eb 6d 79 40 b0
0070 db 44 14 fb 9c fc 4d 34 05 7f ad ab f4 ce 36 7c
0080 1b d4 18 f1 f4 b3 f5 25 95 0c a2 21 15 93 21 13
0090 d6 9e ec 48 5b d1 cc 83 03 b6 c4 8b ab 0b c0 ee
00a0 35 f7 d5 e2 ad 6e 7d 5b d1 a2 50 2b 44 31 ab 36
00b0 cf f0 93 2d 95 ca a2 a0 70 d8 69 f0 1e f0 3a 64
00c0 be 2b 4e 5a 49 9c 42 f3 4d 51 f9 71 f6 10 a4 f8
00d0 d6 04 d0 14 7b
Frame (213 bytes)
Decrypted TLS (137 bytes)
Ethernet (eth), 14 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default

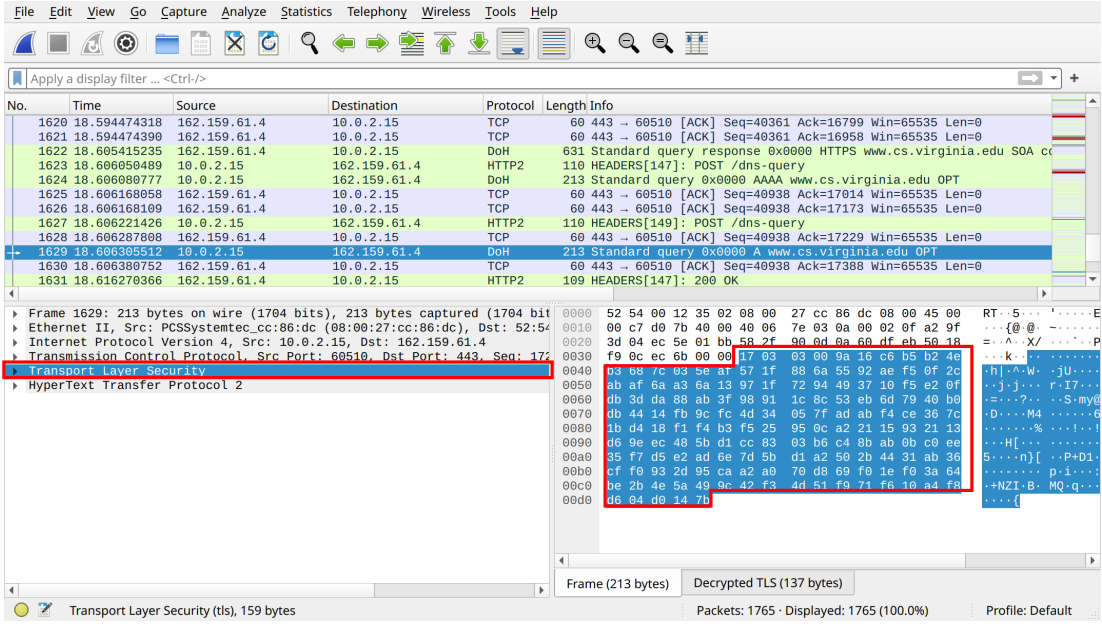

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Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
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1621 18.594474390 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16958 Win=65535 Len=0
1622 18.605415235 162.159.61.4 10.0.2.15 DoH 631 Standard query response 0x0000 HTTPS www.cs.virginia.edu SOA c
1623 18.606050489 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[147]: POST /dns-query
1624 18.606080777 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 AAAA www.cs.virginia.edu OPT
1625 18.606168058 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17014 Win=65535 Len=0
1626 18.606168109 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17173 Win=65535 Len=0
1627 18.606221426 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[149]: POST /dns-query
1628 18.606287808 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17229 Win=65535 Len=0
1629 18.606305512 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 A www.cs.virginia.edu OPT
1630 18.606380752 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17388 Win=65535 Len=0
1631 18.616270366 162.159.61.4 10.0.2.15 HTTP2 109 HEADERS[147]: 200 OK
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 bit
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:54
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 172
Transport Layer Security
HyperText Transfer Protocol 2
0000 52 54 00 12 35 02 08 00 27 cc 86 dc 08 00 45 00
0010 00 c7 d0 7b 40 00 40 06 7e 03 0a 00 02 0f a2 9f
0020 3d 04 ec 5e 01 bb 58 2f 90 0d 0a 60 df eb 50 18
0030 f9 0c ec 6b 00 00 17 03 03 00 9a 16 c6 b5 b2 4e
0040 b3 68 7c 03 5e af 57 1f 88 6a 55 92 ae f5 0f 2c
0050 ab af 6a a3 6a 13 97 1f 72 94 49 37 10 f5 e2 0f
0060 db 3d da 88 ab 3f 98 91 1c 8c 53 eb 6d 79 40 b0
0070 db 44 14 fb 9c fc 4d 34 05 7f ad ab f4 ce 36 7c
0080 1b d4 18 f1 f4 b3 f5 25 95 0c a2 21 15 93 21 13
0090 d6 9e ec 48 5b d1 cc 83 03 b6 c4 8b ab 0b c0 ee
00a0 35 f7 d5 e2 ad 6e 7d 5b d1 a2 50 2b 44 31 ab 36
00b0 cf f0 93 2d 95 ca a2 a0 70 d8 69 f0 1e f0 3a 64
00c0 be 2b 4e 5a 49 9c 42 f3 4d 51 f9 71 f6 10 a4 f8
00d0 d6 04 d0 14 7b
Frame (213 bytes)
Decrypted TLS (137 bytes)
Ethernet (eth), 14 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default

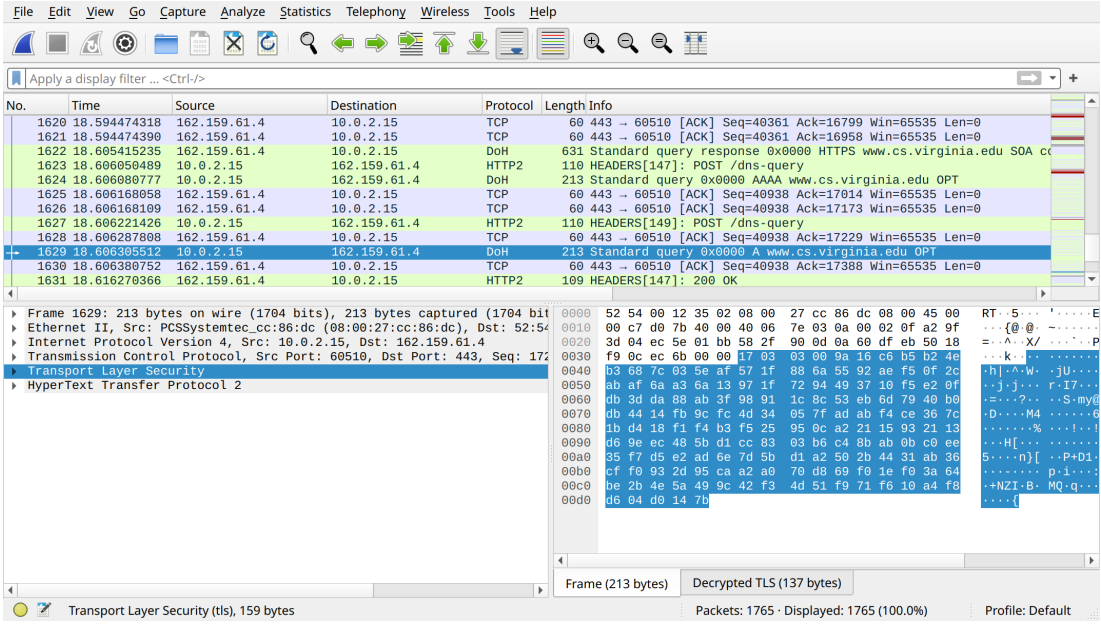

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1620 18.594474318 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16799 Win=65535 Len=0
1621 18.594474390 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16958 Win=65535 Len=0
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1625 18.606168058 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17014 Win=65535 Len=0
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1627 18.606221426 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[149]: POST /dns-query
1628 18.606287808 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17229 Win=65535 Len=0
1629 18.606305512 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 A www.cs.virginia.edu OPT
1630 18.606380752 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17388 Win=65535 Len=0
1631 18.616270366 162.159.61.4 10.0.2.15 HTTP2 109 HEADERS[147]: 200 OK
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 bit
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:54
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 172
Transport Layer Security
HyperText Transfer Protocol 2
0000 52 54 00 12 35 02 08 00 27 cc 86 dc 08 00 45 00
0010 00 c7 d0 7b 40 00 40 06 7e 03 0a 00 02 0f a2 9f
0020 3d 04 ec 5e 01 bb 58 2f 90 0d 0a 60 df eb 50 18
0030 f9 0c ec 6b 00 00 17 03 03 00 9a 16 c6 b5 b2 4e
0040 b3 68 7c 03 5e af 57 1f 88 6a 55 92 ae f5 0f 2c
0050 ab af 6a a3 6a 13 97 1f 72 94 49 37 10 f5 e2 0f
0060 db 3d da 88 ab 3f 98 91 1c 8c 53 eb 6d 79 40 b0
0070 db 44 14 fb 9c fc 4d 34 05 7f ad ab f4 ce 36 7c
0080 1b d4 18 f1 f4 b3 f5 25 95 0c a2 21 15 93 21 13
0090 d6 9e ec 48 5b d1 cc 83 03 b6 c4 8b ab 0b c0 ee
00a0 35 f7 d5 e2 ad 6e 7d 5b d1 a2 50 2b 44 31 ab 36
00b0 cf f0 93 2d 95 ca a2 a0 70 d8 69 f0 1e f0 3a 64
00c0 be 2b 4e 5a 49 9c 42 f3 4d 51 f9 71 f6 10 a4 f8
00d0 d6 04 d0 14 7b
Frame (213 bytes)
Decrypted TLS (137 bytes)
Internet Protocol Version 4 (ip), 20 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default

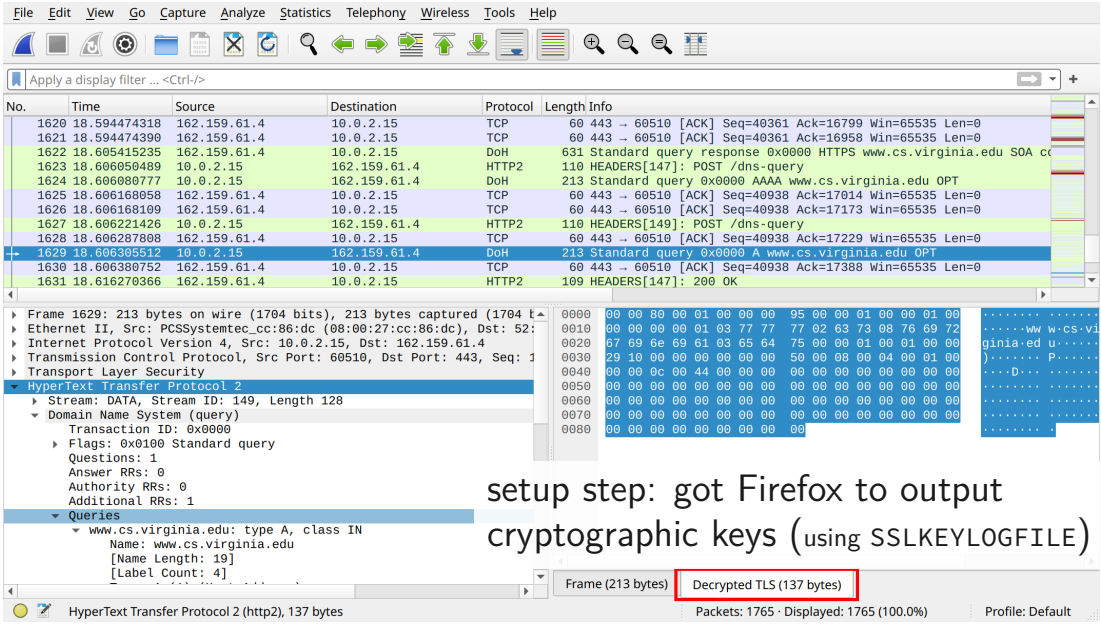

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1620 18.594474318 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16799 Win=65535 Len=0
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1626 18.606168109 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17173 Win=65535 Len=0
1627 18.606221426 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[149]: POST /dns-query
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1630 18.606380752 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17388 Win=65535 Len=0
1631 18.616270366 162.159.61.4 10.0.2.15 HTTP2 109 HEADERS[147]: 200 OK
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 bit
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:54
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 172
Transport Layer Security
HyperText Transfer Protocol 2
0000 52 54 00 12 35 02 08 00 27 cc 86 dc 08 00 45 00
0010 00 c7 d0 7b 40 00 40 06 7e 03 0a 00 02 0f a2 9f
0020 3d 04 ec 5e 01 bb 58 2f 90 0d 0a 60 df eb 50 18
0030 f9 0c ec 6b 00 00 17 03 03 00 9a 16 c6 b5 b2 4e
0040 b3 68 7c 03 5e af 57 1f 88 6a 55 92 ae f5 0f 2c
0050 ab af 6a a3 6a 13 97 1f 72 94 49 37 10 f5 e2 0f
0060 db 3d da 88 ab 3f 98 91 1c 8c 53 eb 6d 79 40 b0
0070 db 44 14 fb 9c fc 4d 34 05 7f ad ab f4 ce 36 7c
0080 1b d4 18 f1 f4 b3 f5 25 95 0c a2 21 15 93 21 13
0090 d6 9e ec 48 5b d1 cc 83 03 b6 c4 8b ab 0b c0 ee
00a0 35 f7 d5 e2 ad 6e 7d 5b d1 a2 50 2b 44 31 ab 36
00b0 cf f0 93 2d 95 ca a2 a0 70 d8 69 f0 1e f0 3a 64
00c0 be 2b 4e 5a 49 9c 42 f3 4d 51 f9 71 f6 10 a4 f8
00d0 d6 04 d0 14 7b
Frame (213 bytes)
Decrypted TLS (137 bytes)
Internet Protocol Version 4 (ip), 20 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
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No. Time Source Destination Protocol Length Info
1620 18.594474318 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16799 Win=65535 Len=0
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1626 18.606168109 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17173 Win=65535 Len=0
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1630 18.606380752 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17388 Win=65535 Len=0
1631 18.616270366 162.159.61.4 10.0.2.15 HTTP2 109 HEADERS[147]: 200 OK
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 bit
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:54
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 172
Transport Layer Security
HyperText Transfer Protocol 2
0000 52 54 00 12 35 02 08 00 27 cc 86 dc 08 00 45 00 RT··5··· '·····E
0010 00 c7 d0 7b 40 00 40 06 7e 03 0a 00 02 0f a2 9f ···{@·@· ~·····
0020 3d 04 ec 5e 01 bb 58 2f 90 0d 0a 60 df eb 50 18 =··^··X/ ···`··P
0030 f9 0c ec 6b 00 00 17 03 03 00 9a 16 c6 b5 b2 4e ···k···· ·····N
0040 b3 68 7c 03 5e af 57 1f 88 6a 55 92 ae f5 0f 2c ·h|·^·W· ·jU····,
0050 ab af 6a a3 6a 13 97 1f 72 94 49 37 10 f5 e2 0f ··j·j··· r·I7····
0060 db 3d da 88 ab 3f 98 91 1c 8c 53 eb 6d 79 40 b0 ·=···?·· ··S·my@·
0070 db 44 14 fb 9c fc 4d 34 05 7f ad ab f4 ce 36 7c ·D····M4 ·····6|
0080 1b d4 18 f1 f4 b3 f5 25 95 0c a2 21 15 93 21 13 ·······% ···!··!·
0090 d6 9e ec 48 5b d1 cc 83 03 b6 c4 8b ab 0b c0 ee ···H[··· ········
00a0 35 f7 d5 e2 ad 6e 7d 5b d1 a2 50 2b 44 31 ab 36 5····n}[··P+D1·6
00b0 cf f0 93 2d 95 ca a2 a0 70 d8 69 f0 1e f0 3a 64 ···-···· p·i···:d
00c0 be 2b 4e 5a 49 9c 42 f3 4d 51 f9 71 f6 10 a4 f8 ·+NZI·B· MQ·q····
00d0 d6 04 d0 14 7b ····{
Frame (213 bytes)
Decrypted TLS (137 bytes)
Transmission Control Protocol (tcp), 20 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default

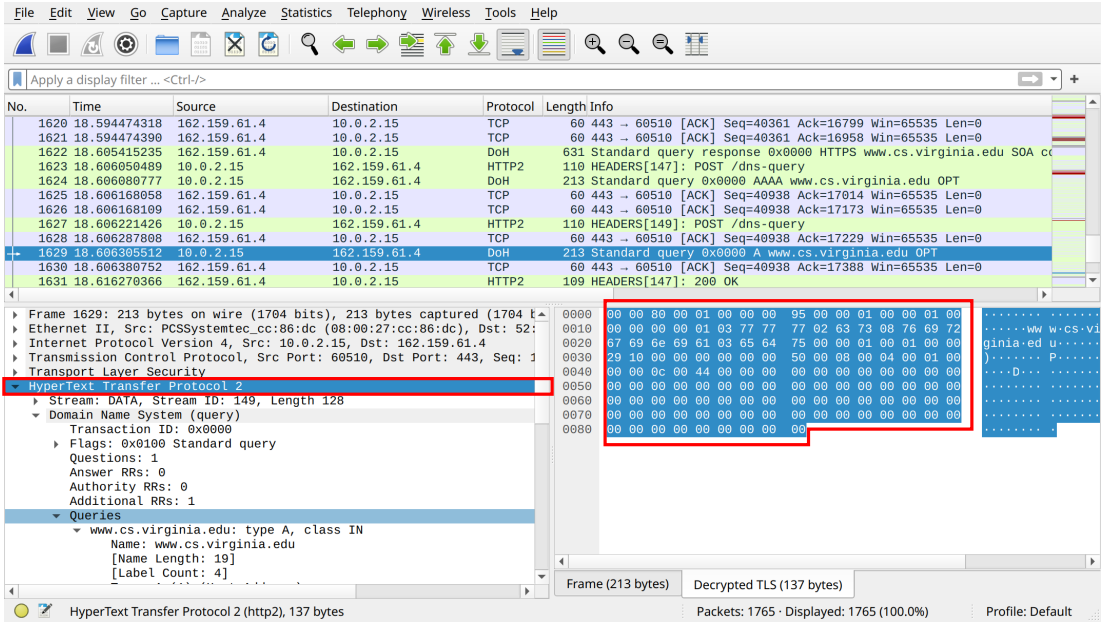

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Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
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1626 18.606168109 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17173 Win=65535 Len=0
1627 18.606221426 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[149]: POST /dns-query
1628 18.606287808 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17229 Win=65535 Len=0
1629 18.606305512 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 A www.cs.virginia.edu OPT
1630 18.606380752 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17388 Win=65535 Len=0
1631 18.616270366 162.159.61.4 10.0.2.15 HTTP2 109 HEADERS[147]: 200 OK
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 bit
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:54
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 172
Transport Layer Security
HyperText Transfer Protocol 2
0000 52 54 00 12 35 02 08 00 27 cc 86 dc 08 00 45 00 RT··5··· '·····E
0010 00 c7 d0 7b 40 00 40 06 7e 03 0a 00 02 0f a2 9f ···{@·@· ~·····
0020 3d 04 ec 5e 01 bb 58 2f 90 0d 0a 60 df eb 50 18 =··^··X/ ···`··P·
0030 f9 0c ec 6b 00 00 17 03 03 00 9a 16 c6 b5 b2 4e ···k···· ·····N
0040 b3 68 7c 03 5e af 57 1f 88 6a 55 92 ae f5 0f 2c ·h|·^·W· ·jU····,
0050 ab af 6a a3 6a 13 97 1f 72 94 49 37 10 f5 e2 0f ··j·j··· r·I7····
0060 db 3d da 88 ab 3f 98 91 1c 8c 53 eb 6d 79 40 b0 ·=···?·· ··S·my@·
0070 db 44 14 fb 9c fc 4d 34 05 7f ad ab f4 ce 36 7c ·D····M4 ·····6|
0080 1b d4 18 f1 f4 b3 f5 25 95 0c a2 21 15 93 21 13 ·······% ···!··!·
0090 d6 9e ec 48 5b d1 cc 83 03 b6 c4 8b ab 0b c0 ee ···H[··· ········
00a0 35 f7 d5 e2 ad 6e 7d 5b d1 a2 50 2b 44 31 ab 36 5····n}[··P+D1·6
00b0 cf f0 93 2d 95 ca a2 a0 70 d8 69 f0 1e f0 3a 64 ···-···· p·i···:d
00c0 be 2b 4e 5a 49 9c 42 f3 4d 51 f9 71 f6 10 a4 f8 ·+NZI·B· MQ·q····
00d0 d6 04 d0 14 7b ····{
Frame (213 bytes)
Decrypted TLS (137 bytes)
Transmission Control Protocol (tcp), 20 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default

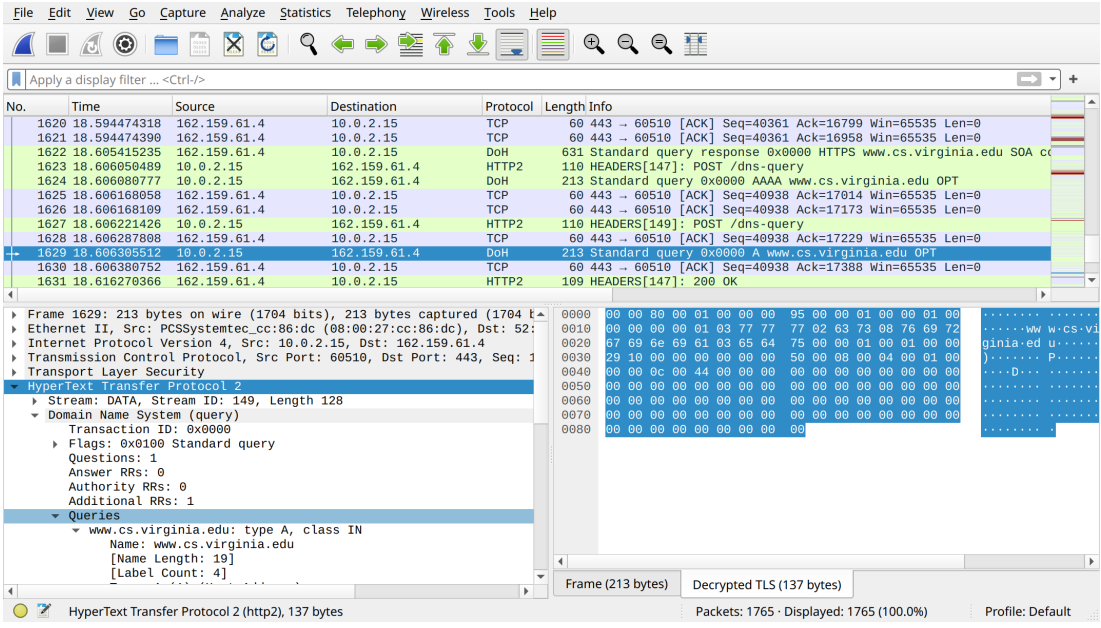

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1620 18.594474318 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16799 Win=65535 Len=0
1621 18.594474390 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16958 Win=65535 Len=0
1622 18.605415235 162.159.61.4 10.0.2.15 DoH 631 Standard query response 0x0000 HTTPS www.cs.virginia.edu SOA c
1623 18.606050489 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[147]: POST /dns-query
1624 18.606080777 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 AAAA www.cs.virginia.edu OPT
1625 18.606168058 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17014 Win=65535 Len=0
1626 18.606168109 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17173 Win=65535 Len=0
1627 18.606221426 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[149]: POST /dns-query
1628 18.606287808 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17229 Win=65535 Len=0
1629 18.606305512 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 A www.cs.virginia.edu OPT
1630 18.606380752 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17388 Win=65535 Len=0
1631 18.616270366 162.159.61.4 10.0.2.15 HTTP2 109 HEADERS[147]: 200 OK
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 bit
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:54
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 172
Transport Layer Security
HyperText Transfer Protocol 2
0000 52 54 00 12 35 02 08 00 27 cc 86 dc 08 00 45 00 RT··5··· '·····E
0010 00 c7 d0 7b 40 00 40 06 7e 03 0a 00 02 0f a2 9f ···{@·@· ~·····
0020 3d 04 ec 5e 01 bb 58 2f 90 0d 0a 60 df eb 50 18 =··^··X/ ···`··P
0030 f9 0c ec 6b 00 00 17 03 03 00 9a 16 c6 b5 b2 4e ···k···· ·····N
0040 b3 68 7c 03 5e af 57 1f 88 6a 55 92 ae f5 0f 2c ·h|·^·W· ·jU····,
0050 ab af 6a a3 6a 13 97 1f 72 94 49 37 10 f5 e2 0f ··j·j··· r·I7····
0060 db 3d da 88 ab 3f 98 91 1c 8c 53 eb 6d 79 40 b0 ·=···?·· ··S·my@·
0070 db 44 14 fb 9c fc 4d 34 05 7f ad ab f4 ce 36 7c ·D····M4 ·····6|
0080 1b d4 18 f1 f4 b3 f5 25 95 0c a2 21 15 93 21 13 ·······% ···!··!·
0090 d6 9e ec 48 5b d1 cc 83 03 b6 c4 8b ab 0b c0 ee ···H[··· ·····
00a0 35 f7 d5 e2 ad 6e 7d 5b d1 a2 50 2b 44 31 ab 36 5····n}[··P+D1·6
00b0 cf f0 93 2d 95 ca a2 a0 70 d8 69 f0 1e f0 3a 64 ·····p· ·i···:d
00c0 be 2b 4e 5a 49 9c 42 f3 4d 51 f9 71 f6 10 a4 f8 ·+NZI·B· MQ·q····
00d0 d6 04 d0 14 7b ····{
Frame (213 bytes)
Decrypted TLS (137 bytes)
Transport Layer Security (tls), 159 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default

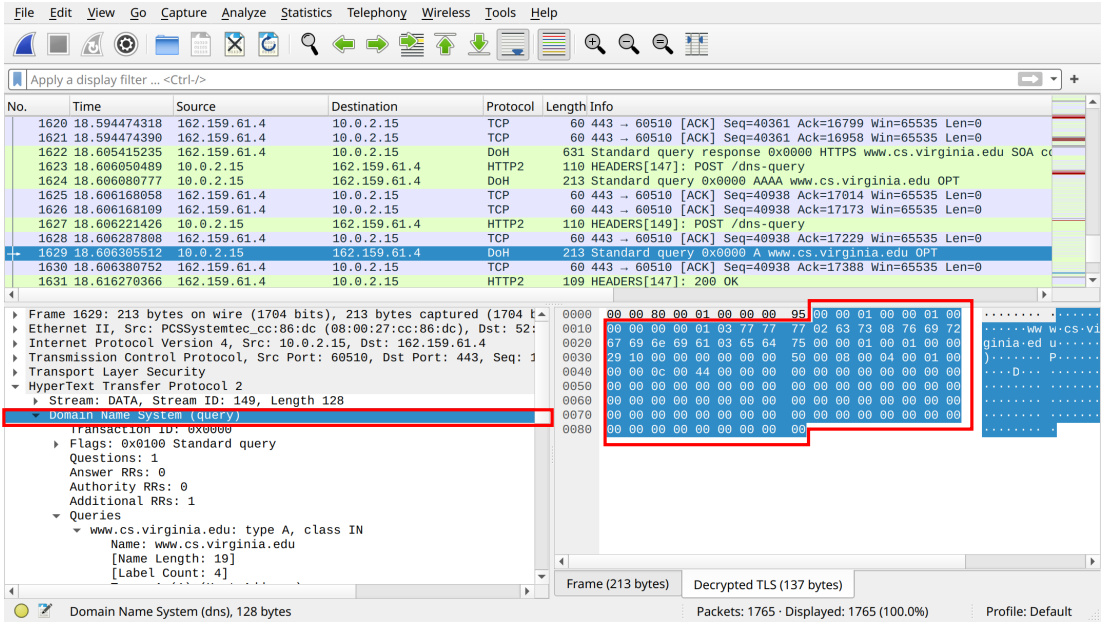

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1620 18.594474318 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16799 Win=65535 Len=0
1621 18.594474390 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16958 Win=65535 Len=0
1622 18.605415235 162.159.61.4 10.0.2.15 DoH 631 Standard query response 0x0000 HTTPS www.cs.virginia.edu SOA c
1623 18.606050489 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[147]: POST /dns-query
1624 18.606080777 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 AAAA www.cs.virginia.edu OPT
1625 18.606168058 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17014 Win=65535 Len=0
1626 18.606168109 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17173 Win=65535 Len=0
1627 18.606221426 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[149]: POST /dns-query
1628 18.606287808 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17229 Win=65535 Len=0
1629 18.606305512 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 A www.cs.virginia.edu OPT
1630 18.606380752 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17388 Win=65535 Len=0
1631 18.616270366 162.159.61.4 10.0.2.15 HTTP2 109 HEADERS[147]: 200 OK
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 bit
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:54
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 172
Transport Layer Security
HyperText Transfer Protocol 2
0000 52 54 00 12 35 02 08 00 27 cc 86 dc 08 00 45 00
0010 00 c7 d0 7b 40 00 40 06 7e 03 0a 00 02 0f a2 9f
0020 3d 04 ec 5e 01 bb 58 2f 90 0d 0a 60 df eb 50 18
0030 f9 0c ec 6b 00 00 17 03 03 00 9a 16 c6 b5 b2 4e
0040 b3 68 7c 03 5e af 57 1f 88 6a 55 92 ae f5 0f 2c
0050 ab af 6a a3 6a 13 97 1f 72 94 49 37 10 f5 e2 0f
0060 db 3d da 88 ab 3f 98 91 1c 8c 53 eb 6d 79 40 b0
0070 db 44 14 fb 9c fc 4d 34 05 7f ad ab f4 ce 36 7c
0080 1b d4 18 f1 f4 b3 f5 25 95 0c a2 21 15 93 21 13
0090 d6 9e ec 48 5b d1 cc 83 03 b6 c4 8b ab 0b c0 ee
00a0 35 f7 d5 e2 ad 6e 7d 5b d1 a2 50 2b 44 31 ab 36
00b0 cf f0 93 2d 95 ca a2 a0 70 d8 69 f0 1e f0 3a 64
00c0 be 2b 4e 5a 49 9c 42 f3 4d 51 f9 71 f6 10 a4 f8
00d0 d6 04 d0 14 7b
Frame (213 bytes)
Decrypted TLS (137 bytes)
Transport Layer Security (tls), 159 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default

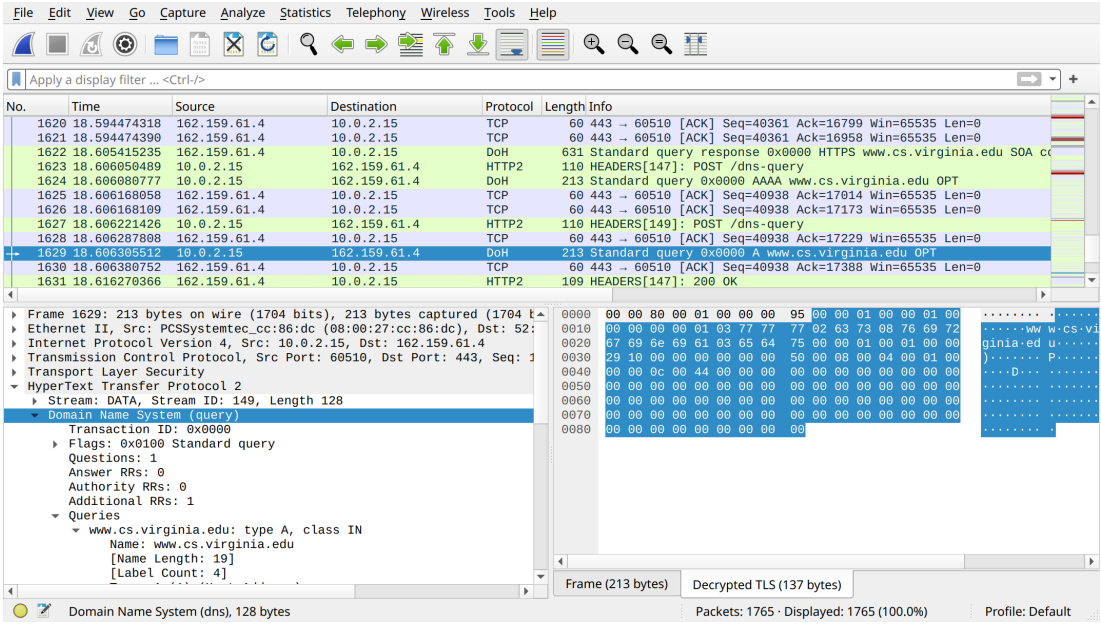


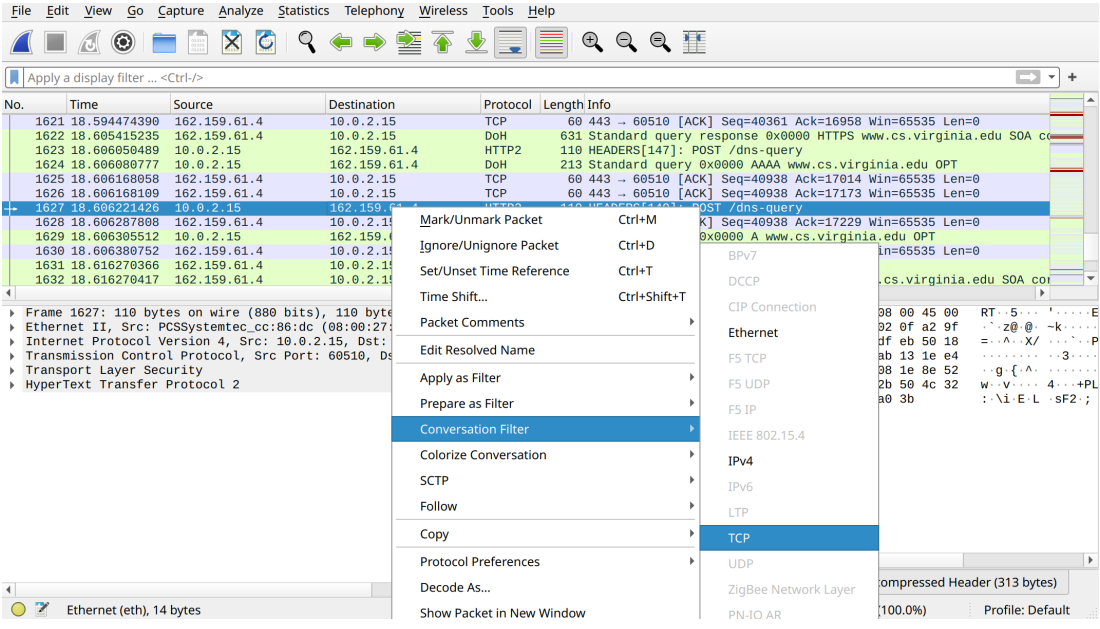
setup step: got Firefox to output cryptographic keys (using SSLKEYLOGFILE)

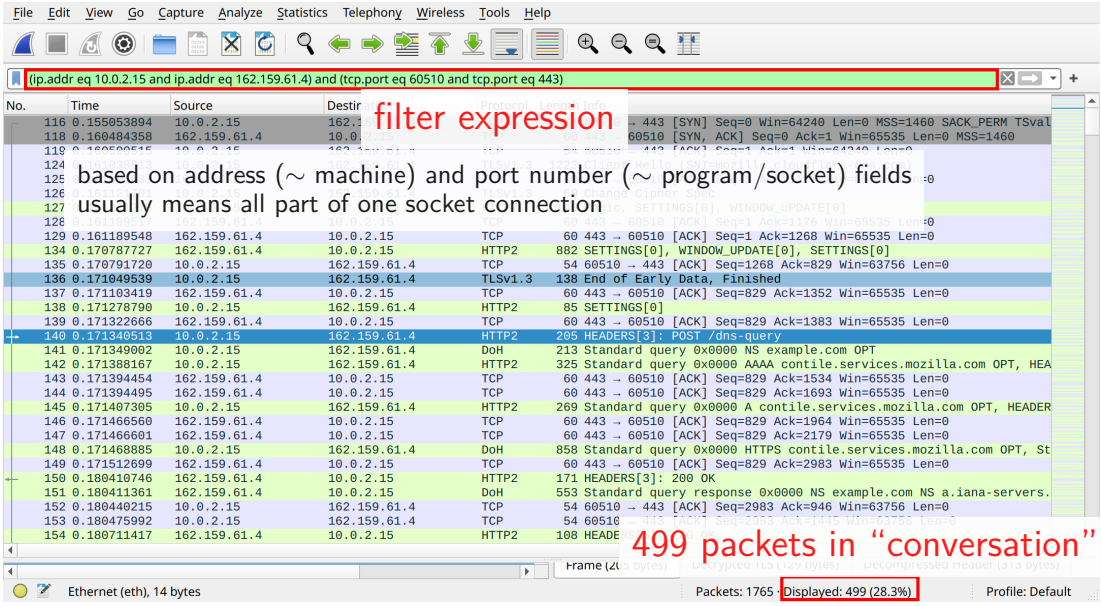

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1620 18.594474318 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16799 Win=65535 Len=0
1621 18.594474390 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16958 Win=65535 Len=0
1622 18.605415235 162.159.61.4 10.0.2.15 DoH 631 Standard query response 0x0000 HTTPS www.cs.virginia.edu SOA c
1623 18.606050489 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[147]: POST /dns-query
1624 18.606080777 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 AAAA www.cs.virginia.edu OPT
1625 18.606168058 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17014 Win=65535 Len=0
1626 18.606168109 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17173 Win=65535 Len=0
1627 18.606221426 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[149]: POST /dns-query
1628 18.606287808 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17229 Win=65535 Len=0
1629 18.606305512 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 A www.cs.virginia.edu OPT
1630 18.606380752 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17388 Win=65535 Len=0
1631 18.616270366 162.159.61.4 10.0.2.15 HTTP2 109 HEADERS[147]: 200 OK
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 b
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 1
Transport Layer Security
HyperText Transfer Protocol 2
Stream: DATA, Stream ID: 149, Length 128
Domain Name System (query)
Transaction ID: 0x0000
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
Queries
www.cs.virginia.edu: type A, class IN
Name: www.cs.virginia.edu
[Name Length: 19]
[Label Count: 4]
0000 00 00 80 00 01 00 00 00 95 00 00 01 00 00 01 00
0010 00 00 00 00 01 03 77 77 77 02 63 73 08 76 69 72
0020 67 69 6e 69 61 03 65 64 75 00 00 01 00 01 00 00
0030 29 10 00 00 00 00 00 00 50 00 08 00 04 00 01 00
0040 00 00 0c 00 44 00 00 00 00 00 00 00 00 00 00 00
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00
Frame (213 bytes)
Decrypted TLS (137 bytes)
HyperText Transfer Protocol 2 (http2), 137 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default

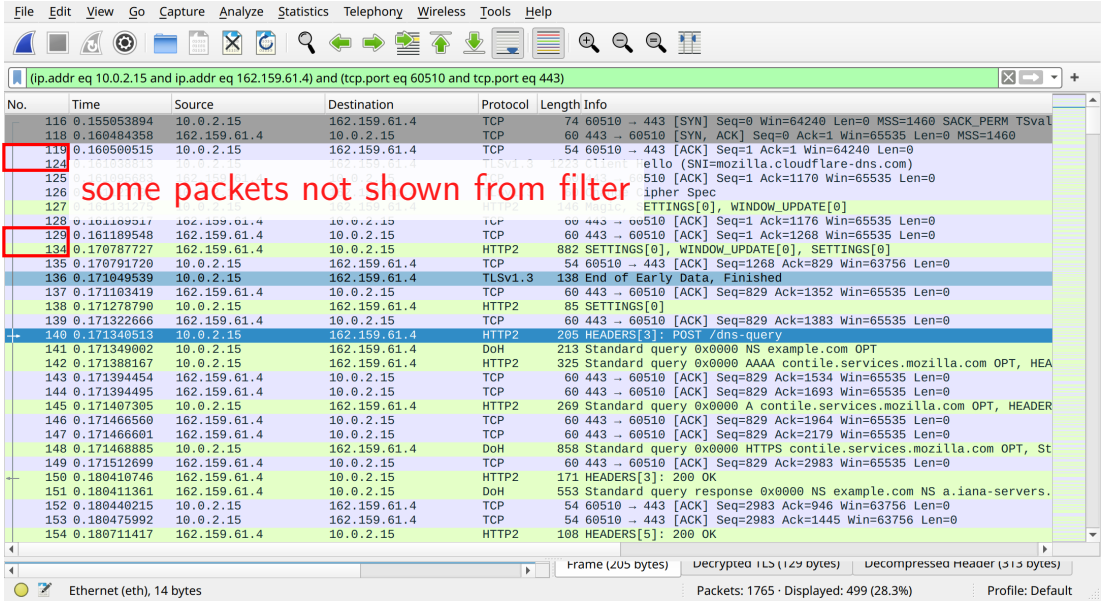

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
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No. Time Source Destination Protocol Length Info
1620 18.594474318 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16799 Win=65535 Len=0
1621 18.594474390 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16958 Win=65535 Len=0
1622 18.605415235 162.159.61.4 10.0.2.15 DoH 631 Standard query response 0x0000 HTTPS www.cs.virginia.edu SOA c
1623 18.606050489 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[147]: POST /dns-query
1624 18.606080777 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 AAAA www.cs.virginia.edu OPT
1625 18.606168058 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17014 Win=65535 Len=0
1626 18.606168109 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17173 Win=65535 Len=0
1627 18.606221426 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[149]: POST /dns-query
1628 18.606287808 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17229 Win=65535 Len=0
1629 18.606305512 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 A www.cs.virginia.edu OPT
1630 18.606380752 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17388 Win=65535 Len=0
1631 18.616270366 162.159.61.4 10.0.2.15 HTTP2 109 HEADERS[147]: 200 OK
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 b
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 1
Transport Layer Security
HyperText Transfer Protocol 2
Stream: DATA, Stream ID: 149, Length 128
Domain Name System (query)
Transaction ID: 0x0000
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
Queries
www.cs.virginia.edu: type A, class IN
Name: www.cs.virginia.edu
[Name Length: 19]
[Label Count: 4]
0000 00 00 80 00 01 00 00 00 95 00 00 01 00 00 01 00
0010 00 00 00 00 01 03 77 77 77 02 63 73 08 76 69 72
0020 67 69 6e 69 61 03 65 64 75 00 00 01 00 01 00 00
0030 29 10 00 00 00 00 00 00 50 00 08 00 04 00 01 00
0040 00 00 0c 00 44 00 00 00 00 00 00 00 00 00 00 00
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00
Frame (213 bytes)
Decrypted TLS (137 bytes)
HyperText Transfer Protocol 2 (http2), 137 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default

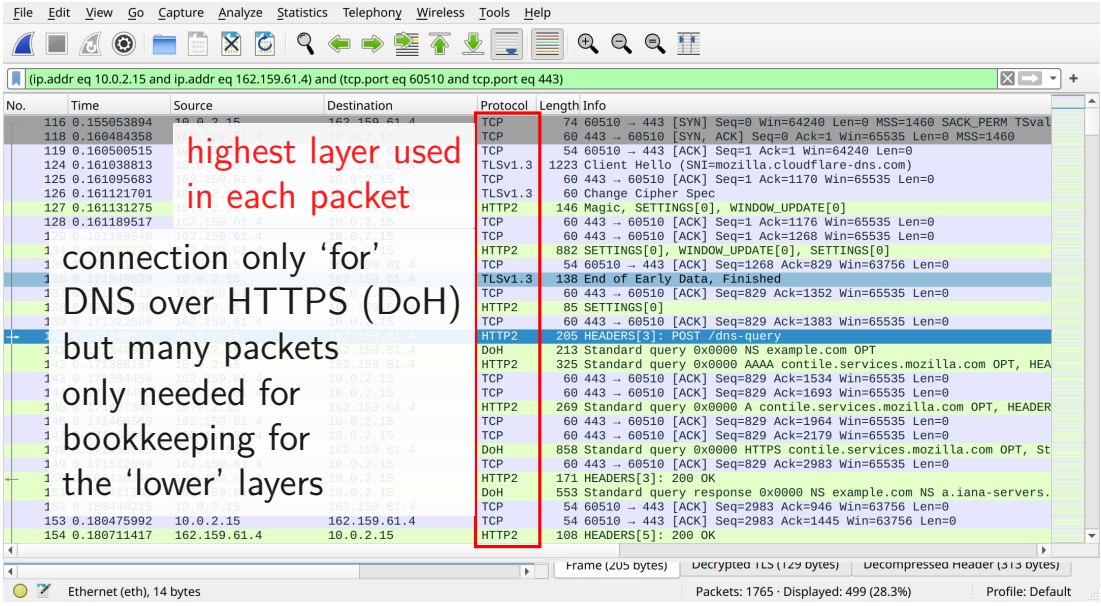

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No. Time Source Destination Protocol Length Info
1620 18.594474318 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16799 Win=65535 Len=0
1621 18.594474390 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16958 Win=65535 Len=0
1622 18.605415235 162.159.61.4 10.0.2.15 DoH 631 Standard query response 0x0000 HTTPS www.cs.virginia.edu SOA c
1623 18.606050489 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[147]: POST /dns-query
1624 18.606080777 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 AAAA www.cs.virginia.edu OPT
1625 18.606168058 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17014 Win=65535 Len=0
1626 18.606168109 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17173 Win=65535 Len=0
1627 18.606221426 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[149]: POST /dns-query
1628 18.606287808 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17229 Win=65535 Len=0
1629 18.606305512 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 A www.cs.virginia.edu OPT
1630 18.606380752 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17388 Win=65535 Len=0
1631 18.616270366 162.159.61.4 10.0.2.15 HTTP2 109 HEADERS[147]: 200 OK
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 b
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 1
Transport Layer Security
HyperText Transfer Protocol 2
Stream: DATA, Stream ID: 149, Length 128
Domain Name System (query)
Transaction ID: 0x0000
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
Queries
www.cs.virginia.edu: type A, class IN
Name: www.cs.virginia.edu
[Name Length: 19]
[Label Count: 4]
0000 00 00 80 00 01 00 00 00 95 00 00 01 00 00 01 00
0010 00 00 00 00 01 03 77 77 77 02 63 73 08 76 69 72
0020 67 69 6e 69 61 03 65 64 75 00 00 01 00 01 00 00
0030 29 10 00 00 00 00 00 00 50 00 08 00 04 00 01 00
0040 00 00 0c 00 44 00 00 00 00 00 00 00 00 00 00 00
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00
Frame (213 bytes)
Decrypted TLS (137 bytes)
Domain Name System (dns), 128 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default

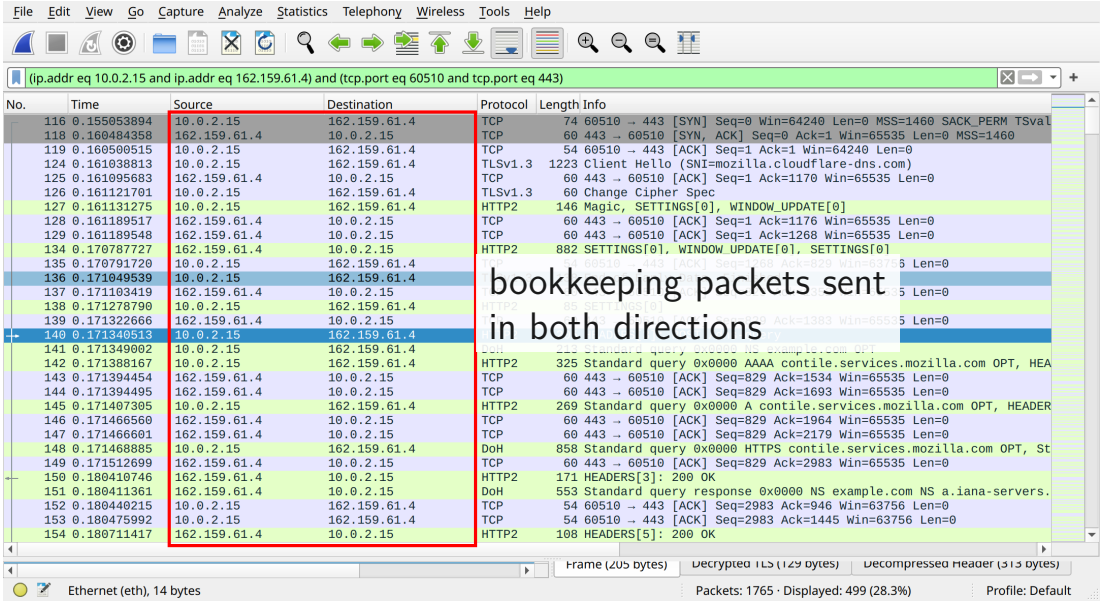

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Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1620 18.594474318 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16799 Win=65535 Len=0
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1629 18.606305512 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 A www.cs.virginia.edu OPT
1630 18.606380752 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17388 Win=65535 Len=0
1631 18.616270366 162.159.61.4 10.0.2.15 HTTP2 109 HEADERS[147]: 200 OK
Frame 1629: 213 bytes on wire (1704 bits), 213 bytes captured (1704 b
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: 52:
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 1
Transport Layer Security
HyperText Transfer Protocol 2
Stream: DATA, Stream ID: 149, Length 128
Domain Name System (query)
Transaction ID: 0x0000
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 1
Queries
www.cs.virginia.edu: type A, class IN
Name: www.cs.virginia.edu
[Name Length: 19]
[Label Count: 4]
0000 00 00 80 00 01 00 00 00 00 95 00 00 01 00 00 01 00
0010 00 00 00 00 01 03 77 77 77 02 63 73 08 76 69 72
0020 67 69 6e 69 61 03 65 64 75 00 00 01 00 01 00 00
0030 29 10 00 00 00 00 00 00 00 50 00 08 00 04 00 01 00
0040 00 00 0c 00 44 00 00 00 00 00 00 00 00 00 00 00
0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0080 00 00 00 00 00 00 00 00
Frame (213 bytes)
Decrypted TLS (137 bytes)
Domain Name System (dns), 128 bytes
Packets: 1765 · Displayed: 1765 (100.0%)
Profile: Default


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1621 18.594474390 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40361 Ack=16958 Win=65535 Len=0
1622 18.605415235 162.159.61.4 10.0.2.15 DoH 631 Standard query response 0x0000 HTTPS www.cs.virginia.edu SOA
1623 18.606050489 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[147]: POST /dns-query
1624 18.606080777 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 AAAA www.cs.virginia.edu OPT
1625 18.606168058 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17014 Win=65535 Len=0
1626 18.606168109 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17173 Win=65535 Len=0
1627 18.606221426 10.0.2.15 162.159.61.4 HTTP2 110 HEADERS[149]: POST /dns-query
1628 18.606287808 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17229 Win=65535 Len=0
1629 18.606305512 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 A www.cs.virginia.edu OPT
1630 18.606380752 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=40938 Ack=17388 Win=65535 Len=0
1631 18.616270366 162.159.61.4 10.0.2.15 DoH 631 Standard query response 0x0000 HTTPS www.cs.virginia.edu SOA
1632 18.616270417 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=41515 Ack=17388 Win=65535 Len=0
Frame 1627: 110 bytes on wire (880 bits), 110 bytes captured (880 bits) on interface eth0, id 0
Ethernet II, Src: PCSSystemtec_cc:86:dc (08:00:27:cc:86:dc), Dst: RealtekU_12:35:02 (52:54:00:12:35:02)
Internet Protocol Version 4, Src: 10.0.2.15, Dst: 162.159.61.4
Transmission Control Protocol, Src Port: 60510, Dst Port: 443, Seq: 16958, Ack: 40361, Len: 44
Transport Layer Security
HyperText Transfer Protocol 2
08 00 45 00 RT··5··· '·····E
92 0f a2 9f ··`·z@·@· ~k·····
df eb 50 18 =···^··X/ ···`··P
ab 13 1e e4 ·····3···
08 1e 8e 52 ··g·{·^· ·····
2b 50 4c 32 w··v···· 4···+PL
a0 3b :·\i·E·L ·sF2·;
Mark/Unmark Packet Ctrl+M
Ignore/Unignore Packet Ctrl+D
Set/Unset Time Reference Ctrl+T
Time Shift... Ctrl+Shift+T
Packet Comments
Edit Resolved Name
Apply as Filter
Prepare as Filter
Conversation Filter
Colorize Conversation
SCTP
Follow
Copy
Protocol Preferences
Decode As...
Show Packet in New Window
BPv7
DCCP
CIP Connection
Ethernet
F5 TCP
F5 UDP
F5 IP
IEEE 802.15.4
IPv4
IPv6
LTP
TCP
UDP
ZigBee Network Layer
PN-IO AR
ompressed Header (313 bytes)
100.0%) Profile: Default
Ethernet (eth), 14 bytes


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
(ip.addr eq 10.0.2.15 and ip.addr eq 162.159.61.4) and (tcp.port eq 60510 and tcp.port eq 443)
No. Time Source Destin Protocol Length Info
filter expression
based on address (∼ machine) and port number (∼ program/socket) fields
usually means all part of one socket connection
116 0.155053894 10.0.2.15 162.
118 0.160484358 162.159.61.4 10.0. 60510 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval
129 0.161189548 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=1 Ack=1268 Win=65535 Len=0
134 0.170787727 162.159.61.4 10.0.2.15 HTTP2 882 SETTINGS[0], WINDOW_UPDATE[0], SETTINGS[0]
135 0.170791720 10.0.2.15 162.159.61.4 TCP 54 60510 → 443 [ACK] Seq=1268 Ack=829 Win=63756 Len=0
136 0.171049539 10.0.2.15 162.159.61.4 TLSv1.3 138 End of Early Data, Finished
137 0.171103419 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=1352 Win=65535 Len=0
138 0.171278790 10.0.2.15 162.159.61.4 HTTP2 85 SETTINGS[0]
139 0.171322666 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=1383 Win=65535 Len=0
140 0.171340513 10.0.2.15 162.159.61.4 HTTP2 205 HEADERS[3]: POST /dns-query
141 0.171349002 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 NS example.com OPT
142 0.171388167 10.0.2.15 162.159.61.4 HTTP2 325 Standard query 0x0000 AAAA contile.services.mozilla.com OPT, HEA
143 0.171394454 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=1534 Win=65535 Len=0
144 0.171394495 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=1693 Win=65535 Len=0
145 0.171407305 10.0.2.15 162.159.61.4 HTTP2 269 Standard query 0x0000 A contile.services.mozilla.com OPT, HEADER
146 0.171466560 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=1964 Win=65535 Len=0
147 0.171466601 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=2179 Win=65535 Len=0
148 0.171468885 10.0.2.15 162.159.61.4 DoH 858 Standard query 0x0000 HTTPS contile.services.mozilla.com OPT, St
149 0.171512699 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=2983 Win=65535 Len=0
150 0.180410746 162.159.61.4 10.0.2.15 HTTP2 171 HEADERS[3]: 200 OK
151 0.180411361 162.159.61.4 10.0.2.15 DoH 553 Standard query response 0x0000 NS example.com NS a.iana-servers.
152 0.180440215 10.0.2.15 162.159.61.4 TCP 54 60510 → 443 [ACK] Seq=2983 Ack=946 Win=63756 Len=0
153 0.180475992 10.0.2.15 162.159.61.4 TCP 54 60510
154 0.180711417 162.159.61.4 10.0.2.15 HTTP2 108 HEADE
499 packets in “conversation”
Ethernet (eth), 14 bytes
Packets: 1765 · Displayed: 499 (28.3%)
Profile: Default


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
(ip.addr eq 10.0.2.15 and ip.addr eq 162.159.61.4) and (tcp.port eq 60510 and tcp.port eq 443)
No. Time Source Destination Protocol Length Info
116 0.155053894 10.0.2.15 162.159.61.4 TCP 74 60510 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval
118 0.160484358 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
119 0.160500515 10.0.2.15 162.159.61.4 TCP 54 60510 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
124 ello (SNI=mozilla.cloudflare-dns.com)
125 510 [ACK] Seq=1 Ack=1170 Win=65535 Len=0
126 ipher Spec
127 ETTINGS[0], WINDOW_UPDATE[0]
128 0.161189517 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=1 Ack=1176 Win=65535 Len=0
129 0.161189548 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=1 Ack=1268 Win=65535 Len=0
134 0.170787727 162.159.61.4 10.0.2.15 HTTP2 882 SETTINGS[0], WINDOW_UPDATE[0], SETTINGS[0]
135 0.170791720 10.0.2.15 162.159.61.4 TCP 54 60510 → 443 [ACK] Seq=1268 Ack=829 Win=63756 Len=0
136 0.171049539 10.0.2.15 162.159.61.4 TLSv1.3 138 End of Early Data, Finished
137 0.171103419 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=1352 Win=65535 Len=0
138 0.171278790 10.0.2.15 162.159.61.4 HTTP2 85 SETTINGS[0]
139 0.171322666 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=1383 Win=65535 Len=0
140 0.171340513 10.0.2.15 162.159.61.4 HTTP2 205 HEADERS[3]: POST /dns-query
141 0.171349002 10.0.2.15 162.159.61.4 DoH 213 Standard query 0x0000 NS example.com OPT
142 0.171388167 10.0.2.15 162.159.61.4 HTTP2 325 Standard query 0x0000 AAAA contile.services.mozilla.com OPT, HEA
143 0.171394454 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=1534 Win=65535 Len=0
144 0.171394495 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=1693 Win=65535 Len=0
145 0.171407305 10.0.2.15 162.159.61.4 HTTP2 269 Standard query 0x0000 A contile.services.mozilla.com OPT, HEADER
146 0.171466560 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=1964 Win=65535 Len=0
147 0.171466601 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=2179 Win=65535 Len=0
148 0.171468885 10.0.2.15 162.159.61.4 DoH 858 Standard query 0x0000 HTTPS contile.services.mozilla.com OPT, St
149 0.171512699 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=2983 Win=65535 Len=0
150 0.180410746 162.159.61.4 10.0.2.15 HTTP2 171 HEADERS[3]: 200 OK
151 0.180411361 162.159.61.4 10.0.2.15 DoH 553 Standard query response 0x0000 NS example.com NS a.iana-servers.
152 0.180440215 10.0.2.15 162.159.61.4 TCP 54 60510 → 443 [ACK] Seq=2983 Ack=946 Win=63756 Len=0
153 0.180475992 10.0.2.15 162.159.61.4 TCP 54 60510 → 443 [ACK] Seq=2983 Ack=1445 Win=63756 Len=0
154 0.180711417 162.159.61.4 10.0.2.15 HTTP2 108 HEADERS[5]: 200 OK
some packets not shown from filter
Frame (205 bytes) Decrypted TLS (129 bytes) Decompressed Header (313 bytes)
Ethernet (eth), 14 bytes
Packets: 1765 · Displayed: 499 (28.3%)
Profile: Default


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
(ip.addr eq 10.0.2.15 and ip.addr eq 162.159.61.4) and (tcp.port eq 60510 and tcp.port eq 443)
No. Time Source Destination Protocol Length Info
116 0.155053894 TCP 74 60510 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval
118 0.160484358 TCP 60 443 → 60510 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
119 0.160500515 TCP 54 60510 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
124 0.161038813 TLSv1.3 1223 Client Hello (SNI=mozilla.cloudflare-dns.com)
125 0.161095683 TCP 60 443 → 60510 [ACK] Seq=1 Ack=1170 Win=65535 Len=0
126 0.161121701 TLSv1.3 60 Change Cipher Spec
127 0.161131275 HTTP2 146 Magic, SETTINGS[0], WINDOW_UPDATE[0]
128 0.161189517 TCP 60 443 → 60510 [ACK] Seq=1 Ack=1176 Win=65535 Len=0
TCP 60 443 → 60510 [ACK] Seq=1 Ack=1268 Win=65535 Len=0
HTTP2 882 SETTINGS[0], WINDOW_UPDATE[0], SETTINGS[0]
TCP 54 60510 → 443 [ACK] Seq=1268 Ack=829 Win=63756 Len=0
TLSv1.3 138 End of Early Data, Finished
TCP 60 443 → 60510 [ACK] Seq=829 Ack=1352 Win=65535 Len=0
HTTP2 85 SETTINGS[0]
TCP 60 443 → 60510 [ACK] Seq=829 Ack=1383 Win=65535 Len=0
HTTP2 205 HEADERS[3]: POST /dns-query
DoH 213 Standard query 0x0000 NS example.com OPT
HTTP2 325 Standard query 0x0000 AAAA contile.services.mozilla.com OPT, HEA
TCP 60 443 → 60510 [ACK] Seq=829 Ack=1534 Win=65535 Len=0
TCP 60 443 → 60510 [ACK] Seq=829 Ack=1693 Win=65535 Len=0
HTTP2 269 Standard query 0x0000 A contile.services.mozilla.com OPT, HEADER
TCP 60 443 → 60510 [ACK] Seq=829 Ack=1964 Win=65535 Len=0
TCP 60 443 → 60510 [ACK] Seq=829 Ack=2179 Win=65535 Len=0
DoH 858 Standard query 0x0000 HTTPS contile.services.mozilla.com OPT, St
TCP 60 443 → 60510 [ACK] Seq=829 Ack=2983 Win=65535 Len=0
HTTP2 171 HEADERS[3]: 200 OK
DoH 553 Standard query response 0x0000 NS example.com NS a.iana-servers.
TCP 54 60510 → 443 [ACK] Seq=2983 Ack=946 Win=63756 Len=0
153 0.180475992 10.0.2.15 162.159.61.4 TCP 54 60510 → 443 [ACK] Seq=2983 Ack=1445 Win=63756 Len=0
154 0.180711417 162.159.61.4 10.0.2.15 HTTP2 108 HEADERS[5]: 200 OK
highest layer used in each packet
connection only ‘for’ DNS over HTTPS (DoH) but many packets only needed for bookkeeping for the ‘lower’ layers
Frame (205 bytes)
Ethernet (eth), 14 bytes
Packets: 1765 · Displayed: 499 (28.3%)
Profile: Default


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
(ip.addr eq 10.0.2.15 and ip.addr eq 162.159.61.4) and (tcp.port eq 60510 and tcp.port eq 443)
No. Time Source Destination Protocol Length Info
116 0.155053894 10.0.2.15 162.159.61.4 TCP 74 60510 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval
118 0.160484358 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
119 0.160500515 10.0.2.15 162.159.61.4 TCP 54 60510 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
124 0.161038813 10.0.2.15 162.159.61.4 TLSv1.3 1223 Client Hello (SNI=mozilla.cloudflare-dns.com)
125 0.161095683 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=1 Ack=1170 Win=65535 Len=0
126 0.161121701 10.0.2.15 162.159.61.4 TLSv1.3 60 Change Cipher Spec
127 0.161131275 10.0.2.15 162.159.61.4 HTTP2 146 Magic, SETTINGS[0], WINDOW_UPDATE[0]
128 0.161189517 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=1 Ack=1176 Win=65535 Len=0
129 0.161189548 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=1 Ack=1268 Win=65535 Len=0
134 0.170787727 162.159.61.4 10.0.2.15 HTTP2 882 SETTINGS[0], WINDOW_UPDATE[0], SETTINGS[0]
135 0.170791720 10.0.2.15 162.159.61.4
136 0.171049539 10.0.2.15 162.159.61.4
137 0.171103419 162.159.61.4 10.0.2.15
138 0.171278790 10.0.2.15 162.159.61.4
139 0.171322666 162.159.61.4 10.0.2.15
140 0.171340513 10.0.2.15 162.159.61.4
141 0.171349002 10.0.2.15 162.159.61.4
142 0.171388167 10.0.2.15 162.159.61.4 HTTP2 325 Standard query 0x0000 AAAA contile.services.mozilla.com OPT, HEA
143 0.171394454 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=1534 Win=65535 Len=0
144 0.171394495 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=1693 Win=65535 Len=0
145 0.171407305 10.0.2.15 162.159.61.4 HTTP2 269 Standard query 0x0000 A contile.services.mozilla.com OPT, HEADER
146 0.171466560 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=1964 Win=65535 Len=0
147 0.171466601 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=2179 Win=65535 Len=0
148 0.171468885 10.0.2.15 162.159.61.4 DoH 858 Standard query 0x0000 HTTPS contile.services.mozilla.com OPT, St
149 0.171512699 162.159.61.4 10.0.2.15 TCP 60 443 → 60510 [ACK] Seq=829 Ack=2983 Win=65535 Len=0
150 0.180410746 162.159.61.4 10.0.2.15 HTTP2 171 HEADERS[3]: 200 OK
151 0.180411361 162.159.61.4 10.0.2.15 DoH 553 Standard query response 0x0000 NS example.com NS a.iana-servers.
152 0.180440215 10.0.2.15 162.159.61.4 TCP 54 60510 → 443 [ACK] Seq=2983 Ack=946 Win=63756 Len=0
153 0.180475992 10.0.2.15 162.159.61.4 TCP 54 60510 → 443 [ACK] Seq=2983 Ack=1445 Win=63756 Len=0
154 0.180711417 162.159.61.4 10.0.2.15 HTTP2 108 HEADERS[5]: 200 OK
bookkeeping packets sent
in both directions
Frame (205 bytes)
Ethernet (eth), 14 bytes
Packets: 1765 · Displayed: 499 (28.3%)
Profile: Default

end-to-end argument

Saltzer, Reed, Clark, "End-to-End Arguments in System Design"

"The function in question can completely and correctly be implemented only with the knowledge and help of the application standing at the end points of the communication system. Therefore, providing that questioned function as a feature of the communication system itself is not possible. (Sometimes an incomplete version of the function provided by the communication system may be useful as a performance enhancement.)"

end-to-end argument

Saltzer, Reed, Clark, "End-to-End Arguments in System Design"

"The function in question can completely and correctly be implemented only with the knowledge and help of the application standing at the end points of the communication system. Therefore, providing that questioned function as a feature of the communication system itself is not possible. (Sometimes an incomplete version of the function provided by the communication system may be useful as a performance enhancement.)"

example: reliable file transfer

want to make sure correct data transferred

want to protect against:

- error in hardware/software on sending machine reading file
- bits being flipped in memory on forwarding machine
- communication system flipping bits in data
- hosts crashing during communication

example: reliable file transfer

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want to protect against:

- error in hardware/software on sending machine reading file
- bits being flipped in memory on forwarding machine
- communication system flipping bits in data
- hosts crashing during communication

communication system can't help a lot of these things

example: reliable file transfer

want to make sure correct data transferred

want to protect against:

- error in hardware/software on sending machine reading file
- bits being flipped in memory on forwarding machine
- communication system flipping bits in data
- hosts crashing during communication

communication system can't help a lot of these things

authors experienced router with bad memory/processor

solution: end-to-end checks

want reliable transfer: compare final files (with hash or similar)

"end-to-end" — doesn't care what middle systems do

end-to-end argument

Saltzer, Reed, Clark, "End-to-End Arguments in System Design"

"The function in question can completely and correctly be implemented only with the knowledge and help of the application standing at the end points of the communication system. Therefore, providing that questioned function as a feature of the communication system itself is not possible. (Sometimes an incomplete version of the function provided by the communication system may be useful as a performance enhancement.)"

end-to-end in practice

"narrow waist" of IP doesn't provide many gaurnetees

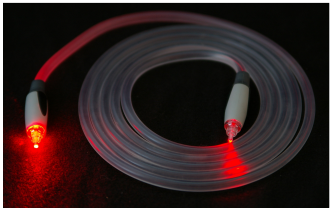
- no gaurentees about reliable transmission, duplicate suppression, message order, ...

but try to provide good service ("best effort")

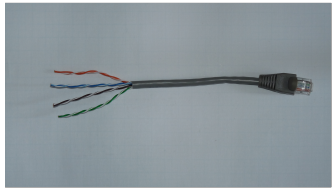
in design: typically middle systems won't know/care about what's forwarded

- but many exceptions

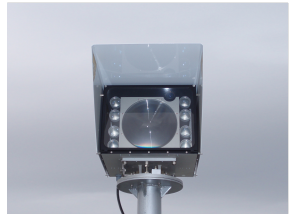
physical media



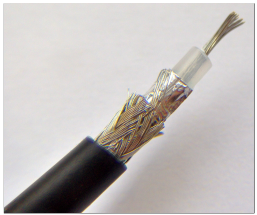
Hustvedt, CC BY-SA 3.0, via Wikimedia Commons



Dmitry G, CC BY-SA 3.0, via Wikimedia Commons



Adamantios, CC BY-SA 3.0, via Wikimedia Commons



FDominec, CC BY-SA 3.0, via Wikimedia Commons

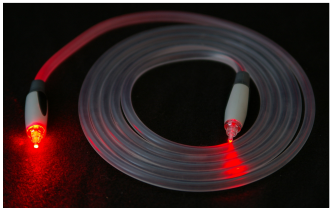


Evan-Amos, via Wikimedia Commons

...

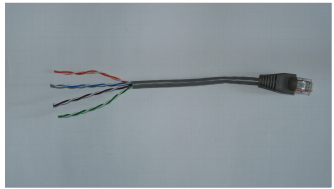
physical media

fiber carrying light



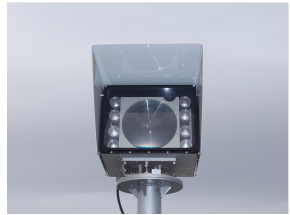
Hustvedt, CC BY-SA 3.0, via Wikimedia Commons

bundle of wires



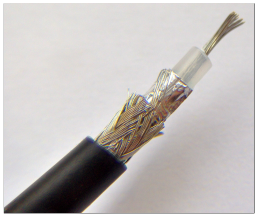
Dmitry G, CC BY-SA 3.0, via Wikimedia Commons

infrared through air



Adamantios, CC BY-SA 3.0, via Wikimedia Commons

single wire



FDominec, CC BY-SA 3.0, via Wikimedia Commons

radio



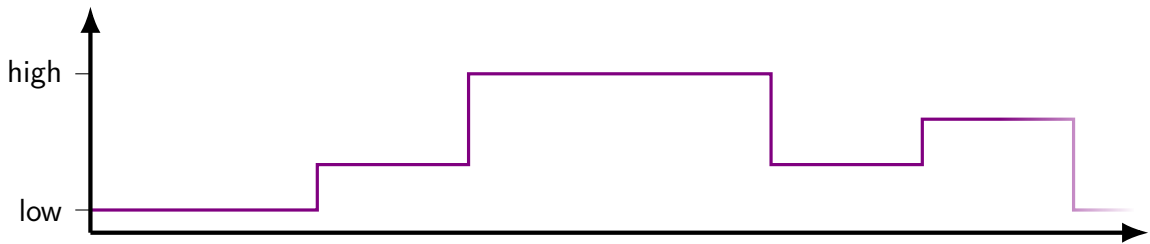
Evan-Amos, via Wikimedia Commons

...

transmitting a signal

can vary...

- voltage
- radio/light intensity
- radio/light frequency
- ...



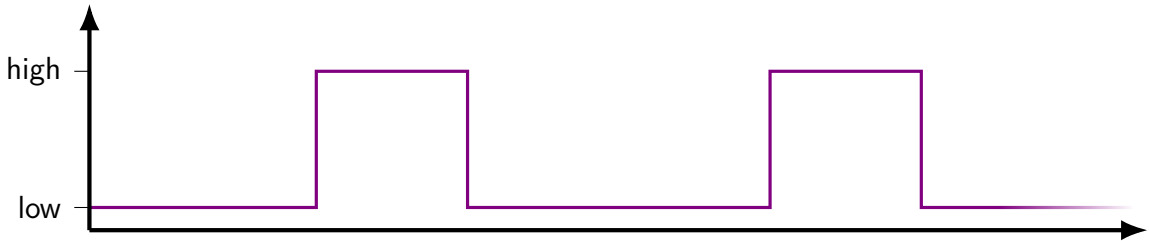
some simplifying assumptions

signal low/high — no in between

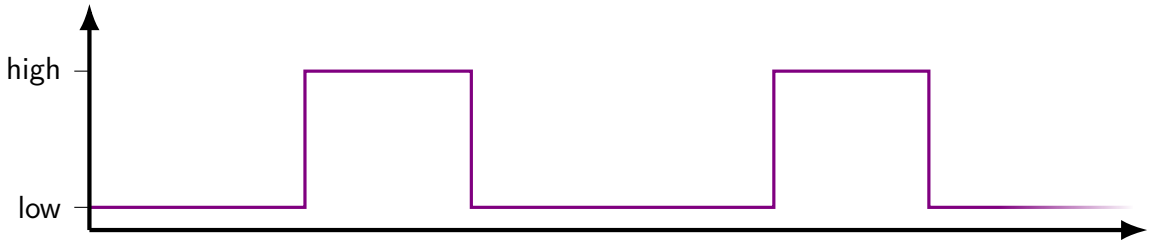
only one 'channel'

- won't have multiple wires/antennas/frequencies/etc.
- won't modulate different things same time

want to send receive bits (0 or 1)



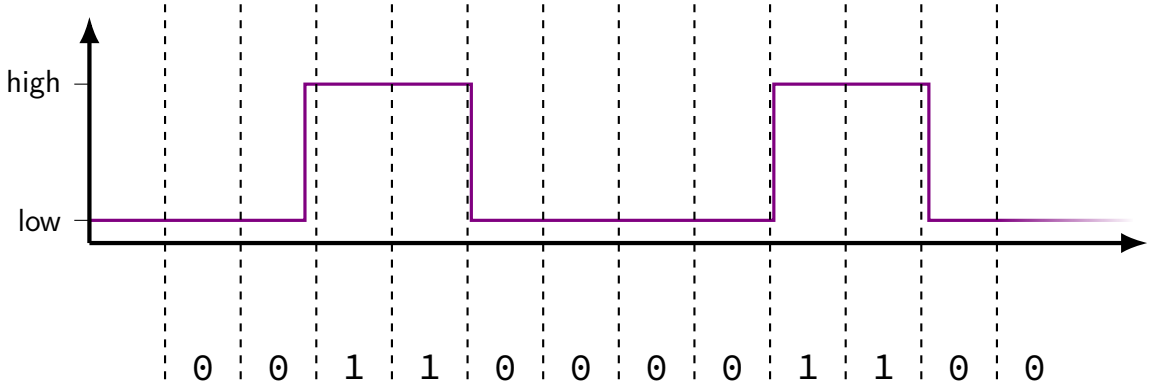
clocking



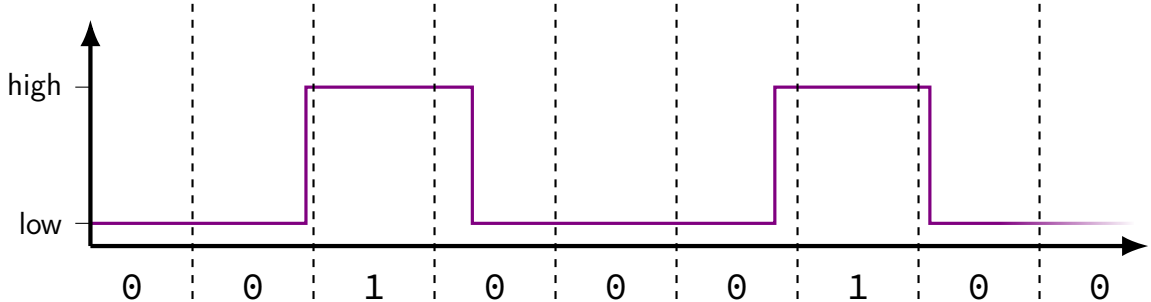
clocking



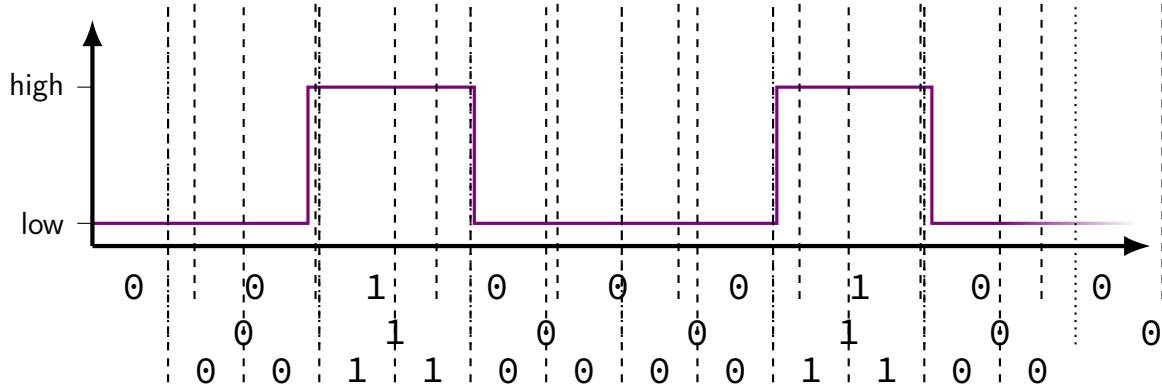
clocking



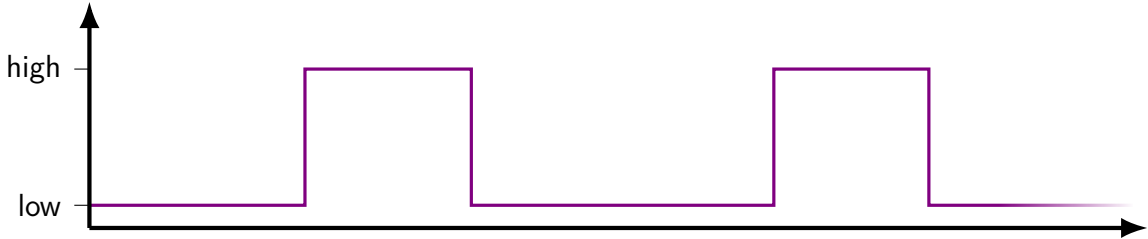
clocking



clocking

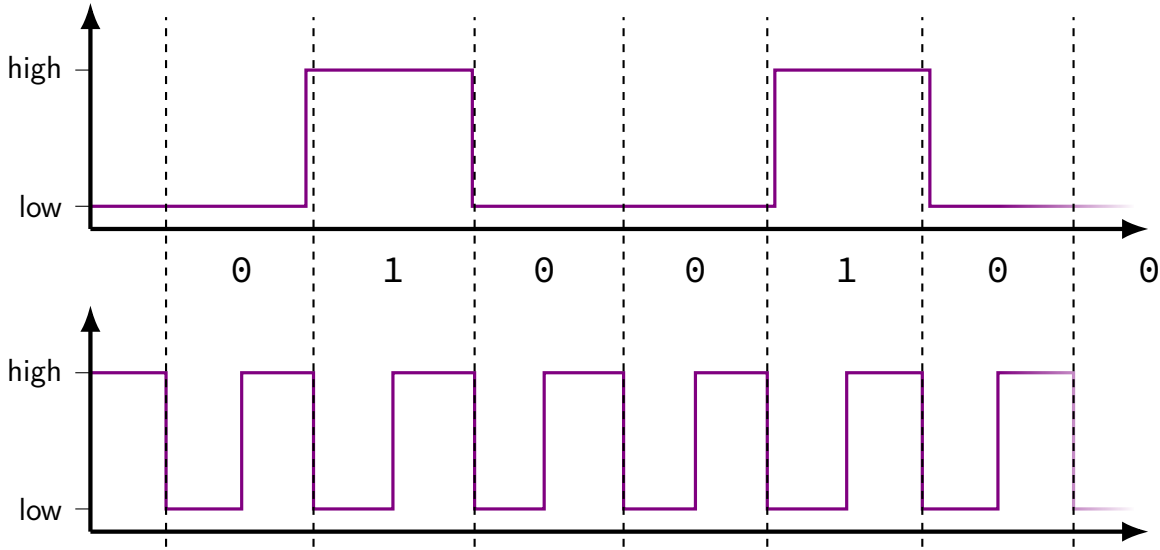


keeping a clock

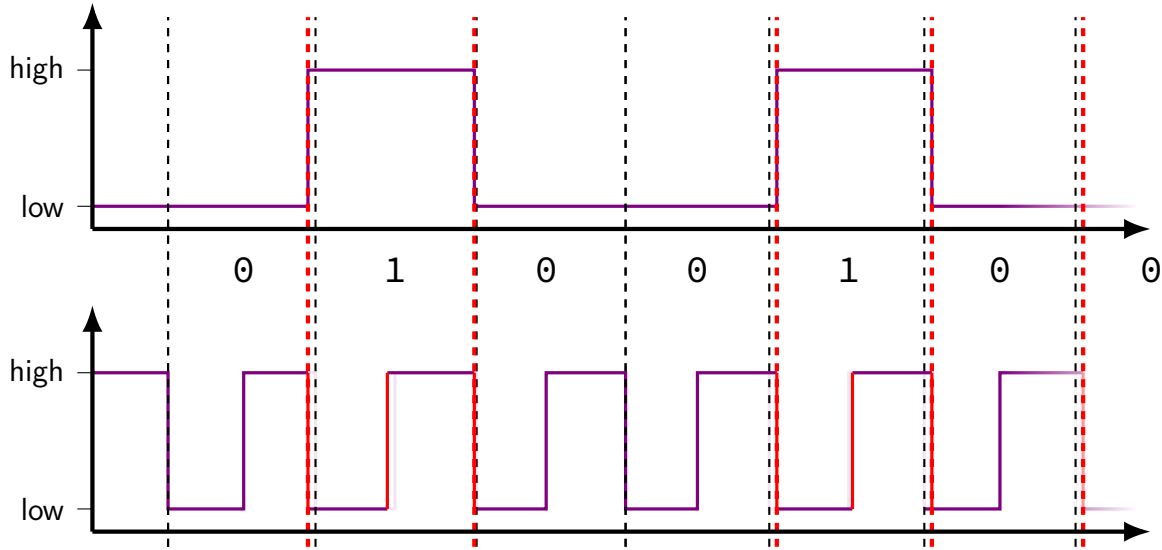




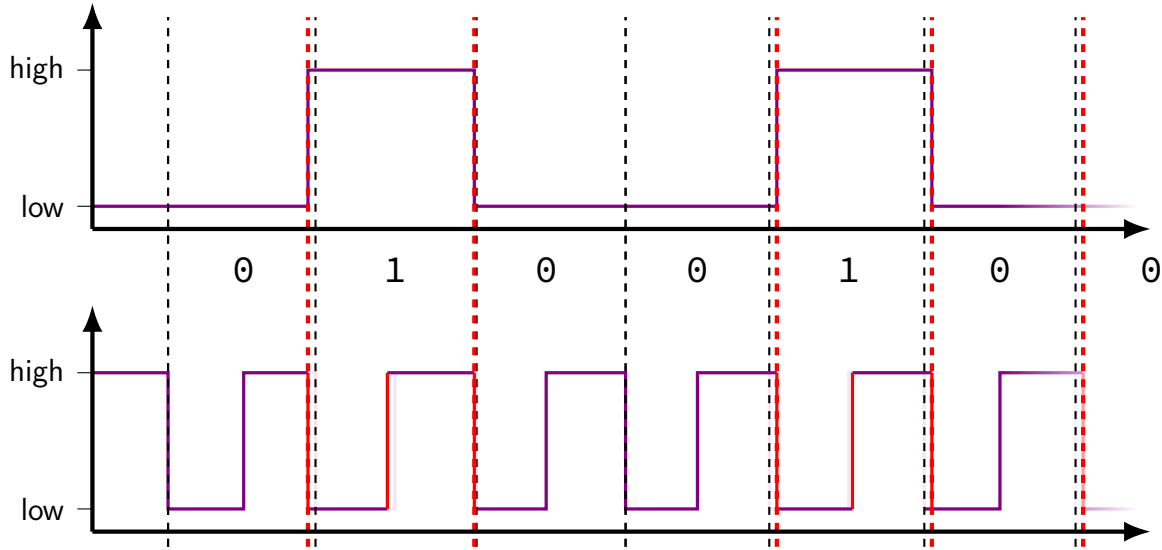
keeping a clock



keeping a clock



keeping a clock



self-synchronizing?

can resynchronize clock by looking for transitions

but doesn't work if lots of consecutive 0s or 1s

also, need to know where low and high point is

important to have transitions to low/high to calibrate this

Manchester encoding

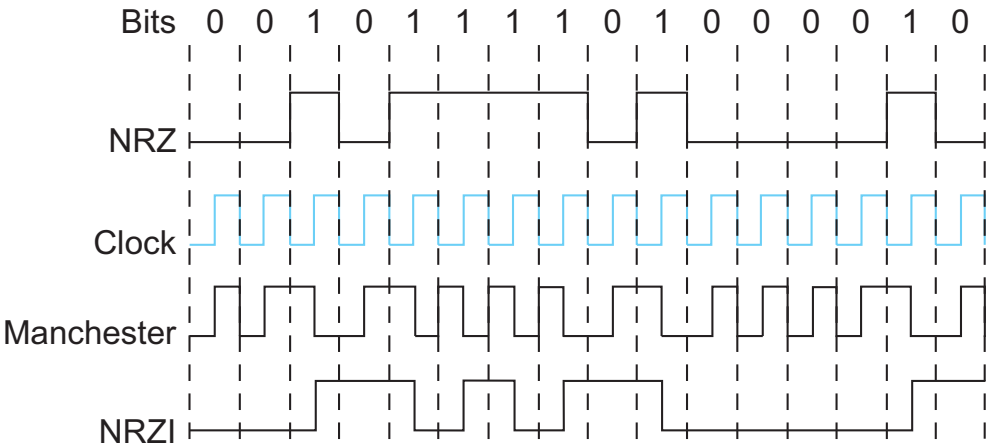


Figure 25 from Section 2.2 of *Computer Networks: A Systems Approach* (6th ed) (Peterson and Davie)

problem with Manchester

fixed the problem of too few transitions

but now most transitions don't send information

means we aren't making goo duse of wire/etc. capacity

there are more clever compromises

(example: 4B5B encoding)

other better encoding options

vary more than just one thing

- example: pulse amplitude and duration

use more than just low/high

...much more

probably covered in ECE Signals course?

backup slides